

SAMSUNG BIOLOGICS CO., LTD.

**Separate Financial Statements
As of and For the Years Ended
December 31, 2016**

ATTACHMENT: INDEPENDENT AUDITORS' REPORT

SAMSUNG BIOLOGICS CO., LTD.

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Independent Auditors' Report

Financial Statements of Samsung Biologics Co., Ltd. as of and for the years ended December 31, 2016 and 2015

Notes to the Financial Statements

Independent Auditors' Report

English Translation of Independent Auditors' Report Originally Issued in Korean on March 16, 2017

To the Board of Directors and Shareholders
Samsung BioLogics Co., Ltd.:

We have audited the accompanying financial statements of Samsung BioLogics Co., Ltd. (the "Company"), which comprise the non-consolidated statement of financial position as of December 31, 2016, and the related non-consolidated statement of comprehensive income, non-consolidated statement of changes in equity and non-consolidated statement of cash flows, all expressed in Korean won, for the year ended December 31, 2016, and a summary of significant accounting policies and other explanatory information.

Management's Responsibility for the Financial Statements

Management is responsible for the preparation and fair presentation of these financial statements in accordance with Korean International Financial Reporting Standards ("K-IFRS") and for such internal control as management determines is necessary to enable the preparation of financial statements that are free from material misstatement, whether due to fraud or error.

Auditors' Responsibility

Our responsibility is to express an audit opinion on these financial statements based on our audit. We conducted our audit in accordance with Korean Standards on Auditing ("KSAs"). Those standards require that we comply with ethical requirements and plan and perform the audit to obtain reasonable assurance about whether the financial statements are free from material misstatement.

An audit involves performing procedures to obtain audit evidence about the amounts and disclosures in the financial statements. The procedures selected depend on the auditor's judgment, including the assessment of the risks of material misstatement of the financial statements, whether due to fraud or error. In making those risk assessments, the auditor considers internal control relevant to the entity's preparation and fair presentation of the financial statements in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of the entity's internal control. An audit also includes evaluating the appropriateness of accounting policies used and the reasonableness of accounting estimates made by management, as well as evaluating the overall presentation of the financial statements.

We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinion.

Opinion

In our opinion, the non-consolidated financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2016, and its financial performance and its cash flows for the year then ended, in accordance with K-IFRS.

Emphasis of Matter

Without qualifying our opinion, we draw attention to the following:

- Loss of control

The Company lost control over a subsidiary in 2015 and does not have any subsidiary in 2016, as described in Note 1. Accordingly, the comparative financial statements as of December 31, 2015, are non-consolidated financial statements.

Other Matter

The financial statements of the Company for the year ended December 31, 2015, were audited by other auditor, who expressed an unqualified opinion on those financial statements on March 16, 2016. The non-consolidated statement of financial position as of December 31, 2015, which is comparatively presented, does not differ, in all material respects, from such audited statement of financial position.

March 16, 2017

Notice to Readers

This report is effective as of March 16, 2017, the auditors' report date. Certain subsequent events or circumstances may have occurred between the auditors' report date and the time the auditors' report is read. Such events or circumstances could significantly affect the accompanying non-consolidated financial statements and may result in modifications to the auditors' report.

SAMSUNG BIOLOGICS CO., LTD.

SEPARATE FINANCIAL STATEMENTS
AS OF AND FOR THE YEARS ENDED
DECEMBER 31, 2016 AND 2015

The accompanying financial statements, including all footnote disclosures, were prepared by, and are the responsibility of, Samsung BioLogics Co., Ltd.

Tae Han Kim
Representative Director
SAMSUNG BIOLOGICS CO., LTD.

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF FINANCIAL POSITION

AS OF DECEMBER 31, 2016 AND 2015

	Note	December 31, 2016	December 31, 2015
		(In thousands of Korean won)	
<u>ASSETS</u>			
CURRENT ASSETS:			
Cash and cash equivalents	5,6,30	₩ 230,337,601	₩ 34,510,062
Short-term financial instruments	5,30	970,000,000	-
Trade and other receivables	5,8,28,30	75,630,820	37,009,243
Inventories	9	164,219,980	119,177,340
Other financial assets	5,10,30	16,289,300	-
Other current assets	10	4,947,475	2,157,249
Total current assets		1,461,425,176	192,853,894
NON-CURRENT ASSETS:			
Other financial assets	5,7,30	8,000	8,000
Investments in associates and joint Ventures	1,11,27,28	4,944,356,290	4,837,539,391
Property, plant and equipment	12,28	1,090,959,766	890,256,137
Intangible assets	13,28	14,044,654	14,285,740
Other non-current assets	5,10,30	22,222,220	25,550,553
Total non-current assets		6,071,590,930	5,767,639,821
TOTAL ASSETS		₩ 7,533,016,106	₩ 5,960,493,715
LIABILITIES AND SHAREHOLDERS' EQUITY			
CURRENT LIABILITIES:			
Trade and other payables	5,15,28,30	₩ 138,056,375	₩ 56,193,891
Financial liabilities at fair value through profit or loss	5,27,30	1,874,470,943	1,820,439,050
Short-term borrowings	5,14,27,30	-	10,109,490
Current portion of long-term borrowings	5,14,30	403,694,084	-
Other current liabilities	16,30	60,323,791	24,926,773
Total current liabilities		2,476,545,193	1,911,669,204
NON-CURRENT LIABILITIES:			
Trade and other payables	5,15,30	2,300,380	1,909,574
Long-term borrowings	5,14,27,30	439,363,194	681,792,000
Defined benefit obligations	17	5,757,265	8,714,355
Deferred tax liabilities	25	526,670,614	581,574,424
Total non-current liabilities		974,091,453	1,273,990,353
TOTAL LIABILITIES		₩ 3,450,636,646	₩ 3,185,659,557

(Continued)

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF FINANCIAL POSITION (CONTINUED)

AS OF DECEMBER 31, 2016 AND 2015

	Note	December 31, 2016	December 31, 2015
		(In thousands of Korean won)	
<u>LIABILITIES AND SHAREHOLDERS' EQUITY</u>			
SHAREHOLDERS' EQUITY:			
Share capital	18	₩ 165,412,500	₩ 137,843,605
Share premium	18	2,487,313,082	1,030,620,604
Accumulated other comprehensive income	18	4,947,005	4,839,666
Retained earnings		<u>1,424,706,873</u>	<u>1,601,530,283</u>
Total shareholders' equity		<u>4,082,379,460</u>	<u>₩ 2,774,834,158</u>
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY		<u>₩ 7,533,016,106</u>	<u>₩ 5,960,493,715</u>
(Concluded)			

See notes

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF COMPREHENSIVE INCOME
FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015

	Note	2016	2015
		(In thousands of Korean won)	
REVENUE	4,19,28	₩ 294,622,021	₩ 91,278,029
COST OF SALES	4,9,20,22,28	<u>268,088,358</u>	<u>114,784,617</u>
GROSS PROFIT (LOSS)		26,533,663	(23,506,588)
Selling, general and administrative expenses	4,21,22,28	<u>56,956,302</u>	<u>180,135,856</u>
OPERATING LOSS		(30,422,639)	(203,642,444)
Other income	23,28	1,217,830	2,663,581
Other expenses	23	(39,010)	(821,253)
Financial income	24	14,348,567	6,844,724
Financial expenses	12,24	(86,783,081)	(1,841,196,798)
Gain on disposal of investment in subsidiary		-	4,543,610,535
Share of loss of equity-accounted investees	11	(129,222,650)	(19,395,735)
PROFIT (LOSS) BEFORE INCOME TAX		(230,900,983)	2,488,062,610
INCOME TAX EXPENSE	25	(54,077,573)	583,116,693
NET PROFIT (LOSS) FOR THE PERIOD			
ATTRIBUTABLE TO:		₩ (176,823,410)	₩ 1,904,945,917
Owners of the Company		(176,823,410)	1,920,178,774
Non-controlling interests		-	(15,232,857)
OTHER COMPREHENSIVE PROFIT (LOSS)		<u>₩ 107,338</u>	<u>₩ (8,535,493)</u>
ITEMS THAT WILL NEVER BE RECLASSIFIED TO PROFIT (LOSS):			
Remeasurement of defined benefit plans	17,25	(1,595,928)	(2,971,127)
Net change in other comprehensive income of equity-accounted investees	11,25	(1,001,039)	-
ITEMS THAT ARE OR MAY BE RECLASSIFIED TO PROFIT (LOSS):			
Net change in other comprehensive income of equity-accounted investees	11	2,704,305	(5,565,214)
Foreign currency translation differences for foreign operations		-	848
TOTAL COMPREHENSIVE PROFIT (LOSS) FOR THE PERIOD		<u>₩ (176,716,072)</u>	<u>₩ 1,896,410,424</u>

(Continued)

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF COMPREHENSIVE INCOME (CONTINUED)

FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015

	Note	2016	2015
		(In thousands of Korean won)	
TOTAL COMPREHENSIVE PROFIT (LOSS)			
ATTRIBUTABLE TO:			
Owners of the Company		(176,716,072)	1,911,863,952
Non-controlling interests		-	(15,453,529)
TOTAL COMPREHENSIVE PROFIT (LOSS)			
FOR THE PERIOD		<u>(176,716,072)</u>	<u>1,896,410,424</u>
EARNINGS (LOSS) PER SHARE			
Basic earnings (loss) per share (in Korean won)	26	<u>₩ (3,115)</u>	<u>₩ 38,828</u>
(Concluded)			

See notes

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF CHANGES IN EQUITY

FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015

	Share capital	Share premium	Accumulated other comprehensive income	Retained earnings (accumulated deficit)	Non-controlling Interests	Total equity
	(In thousands of Korean won)					
Balance at January 1, 2015	₩ 101,833,245	₩ 801,800,313	₩ 9,170,941	₩ (314,664,171)	₩ 32,591,919	₩ 630,732,247
Comprehensive profit (loss) for the period:						
Net profit (loss) for the period	-	-	-	1,920,178,774	(15,232,857)	1,904,945,917
Remeasurement of defined benefit plans	-	-	(2,750,381)	-	(220,746)	(2,971,127)
Net change in other comprehensive income of equity-accounted investees	-	-	(5,565,214)	-	-	(5,565,214)
Foreign currency translation differences for foreign operations	-	-	773	-	75	848
Transactions with owners of the Company recognized directly in equity:						
Capital increase	36,010,360	231,016,681	-	-	-	267,027,041
Owners of the company's capital increase	-	-	-	-	6,228,425	6,228,425
Changes in equity of subsidiary	-	(2,196,389)	-	-	2,196,389	-
Loss of control over a subsidiary	-	-	3,983,547	(3,984,320)	(25,563,205)	(25,563,978)
Balance at December 31, 2015	<u>₩ 137,843,605</u>	<u>₩ 1,030,620,605</u>	<u>₩ 4,839,666</u>	<u>₩ 1,601,530,283</u>	<u>₩ -</u>	<u>₩ 2,774,834,159</u>
Balance at January 1, 2016	₩ 137,843,605	₩ 1,030,620,605	₩ 4,839,666	₩ 1,601,530,283	₩ -	₩ 2,774,834,159
Comprehensive profit (loss) for the period:						
Net profit (loss) for the period	-	-	-	(176,823,410)	-	(176,823,410)
Remeasurement of defined benefit plans	-	-	(1,595,928)	-	-	(1,595,928)
Net change in other comprehensive income of equity-accounted investees	-	-	1,703,267	-	-	1,703,267
Transactions with owners of the Company recognized directly in equity:						
Capital increase	27,568,895	1,456,692,477	-	-	-	1,484,261,372
Balance at December 31, 2016	<u>₩ 165,412,500</u>	<u>₩ 2,487,313,082</u>	<u>₩ 4,947,005</u>	<u>₩ 1,424,706,873</u>	<u>₩ -</u>	<u>₩ 4,082,379,460</u>

See notes

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF CASH FLOWS

FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015

	2016	2015
	(In thousands of Korean won)	
CASH FLOWS FROM OPERATING ACTIVITIES:		
Net profit (loss) for the period	₩ (176,823,410)	₩ 1,904,945,917
Adjustments for:		
Retirement benefits	6,377,009	5,894,646
Depreciation	55,784,426	32,694,637
Amortization	4,365,463	8,933,636
Reversal of loss on valuation of inventories	(1,275,392)	(5,391,777)
Loss on inventories abandoned	321,930	141,983
Loss on disposal of property, plant and equipment	1,033	583,987
Gain on disposal of intangible assets	-	(682,926)
Gain on foreign currency translation	(2,825,232)	(554,928)
Loss on foreign currency translation	3,962,749	9,188,935
Loss on valuation of derivatives	54,031,894	1,820,439,050
Gain on disposal of investment in subsidiary	-	(4,543,610,535)
Gain from assets contributed	(12,018)	-
Interest income	(3,639,650)	(1,182,311)
Interest expenses	19,254,373	2,650,470
Share of loss of equity-accounted investees	129,222,650	19,395,735
Income tax expense	(54,077,573)	583,116,693
	<u>211,491,662</u>	<u>(2,068,382,705)</u>
Change in:		
Decrease (increase) in trade receivables	(28,693,573)	(20,369,472)
Decrease (increase) in other receivables	(4,435,409)	5,804,379
Decrease (increase) in other current assets	(2,502,245)	3,129,492
Increase in inventories	(44,089,178)	(117,238,900)
Increase in long-term other receivables	(12,784,300)	-
Increase in other non-current assets	(359,173)	(18,108,442)
Increase in trade payables	1,161,707	699,431
Increase (decrease) in other payables	(7,659,178)	48,803,685
Increase (decrease) in other current liabilities	35,397,017	6,806,052
Increase in long-term other payables	1,373,929	(577,068)
Decrease in non-current other payables	(526,389)	(1,067,689)
Changes in benefit payment	-	681,446
Payment of plan assets	(10,735,713)	(4,518,212)
	<u>(73,852,505)</u>	<u>(95,955,298)</u>
Interest received	821,632	1,437,110
Interest paid	(24,361,814)	(4,503,598)
Income taxes refund (paid)	63,137	(3,055,177)
	<u>(62,661,298)</u>	<u>(265,513,751)</u>

(Continued)

SAMSUNG BIOLOGICS CO., LTD.

STATEMENTS OF CASH FLOWS (CONTINUED)

FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In thousands of Korean won)	
CASH FLOWS FROM INVESTING ACTIVITIES:		
Cash inflows from investing activities:		
Decrease in long-term financial instruments	₩ -	₩ 1,000
Disposal of property, plant and equipment	-	19,760
Disposal of intangible assets	-	3,500,000
Decrease in deposits provided	-	270,000
Cash outflows for investing activities:		
Increase in short-term financial instruments	970,000,000	-
Acquisition of property, plant and equipment	162,992,350	240,326,345
Acquisition of intangible assets	5,179,126	152,644,276
Acquisition of investment in joint ventures	234,653,000	-
Increase in deposits provided	-	4,832,853
Decrease in loss of control over a subsidiary	-	35,468,801
	<u>(1,372,824,476)</u>	<u>(429,481,515)</u>
CASH FLOWS FROM FINANCING ACTIVITIES		
Cash inflows from financing activities:		
Proceeds from issue of bonds payable	169,743,600	-
Proceeds from short-term borrowings	-	204,786,766
Proceeds from long-term borrowings	28,396,168	248,532,065
Paid in capital increase	1,484,261,372	267,027,041
Cash inflows from changing in capital of subsidiary	-	6,228,425
Cash outflows for financing activities:		
Repayment in short-term borrowings	10,109,490	93,355,664
Repayment in long-term borrowings	41,000,000	-
	<u>1,631,291,650</u>	<u>633,218,633</u>
NET INCREASE(DECREASE) IN CASH AND CASH EQUIVALENTS	195,805,877	(61,776,633)
CASH AND CASH EQUIVALENTS AT THE BEGINNING OF PERIOD	34,510,062	96,286,617
EFFECT OF EXCHANGE RATE CHANGE ON CASH AND CASH EQUIVALENTS	21,662	78
CASH AND CASH EQUIVALENTS AT THE END OF PERIOD	<u><u>₩ 230,337,601</u></u>	<u><u>₩ 34,510,062</u></u>

(Concluded)

See notes

SAMSUNG BIOLOGICS CO., LTD.

NOTES TO FINANCIAL STATEMENTS

AS OF AND FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015,

1. REPORTING ENTITY:

(1) Description of the Company

Samsung BioLogics Co., Ltd. (the “Company”) was established on April 22, 2011, and will be engaged in Biopharmaceutical manufacturing. The Company’s head office is located in 300 Songdo Bio Way (Daero), Yeonsu-gu, Incheon, Korea.

The Company was listed on securities market in November 10, 2016. As of December 31, 2016, the Company's share capital is ₩165,413 million through several capital increases subsequent to establishment.

The shareholders of the Company are as follows:

Shareholder	Shares	Percentage of Ownership (%)
Samsung C&T Corporation	28,742,466	43.44
Samsung Electronics Co., Ltd.	20,836,832	31.49
Quintiles Asia, Inc.	44,400	0.07
Others	16,541,302	25.00
Total	66,165,000	100.00

(2) Subsidiary

The Company lost control over the subsidiary, Samsung Bioepis Co., Ltd., in 2015 and does not have any subsidiary in 2016.

The gain on disposal of investment in subsidiary as the Company lost control over the Samsung Bioepis Co., Ltd. in 2015 is as follows:

	Amounts
	(In thousands of Korean won)
Fair value of acquired shares (A)	₩4,808,578,367
Net assets at the time of loss of control (B)	290,531,810
Non-controlling interests (C)	25,563,205
Items that are or maybe reclassified to profit (loss) (D)	773
Gain on disposal of investment in subsidiary (A-B+C+D)	₩4,543,610,535

As the Company does not have any subsidiary as of December 31, 2016 and 2015, the financial statements as of December 31, 2016 and 2015, are non-consolidated financial statements.

2. BASIS OF ACCOUNTING:

(1) Basis of Preparation

The Company has prepared the financial statements in accordance with the Korean International Financial Reporting Standards (“K-IFRS”).

The principal accounting policies are set out below. Except for the effect of the amendments to K-IFRS and new interpretations set out below, the principal accounting policies used to prepare the financial statements as of and for the year ended December 31, 2016, are consistent with the accounting policies used to prepare the financial statements as of and for the year ended December 31, 2015.

The accompanying financial statements have been prepared on the historical cost basis except for certain non-current assets and financial instruments that are measured at fair values, as explained in the accounting policies below. Historical cost is based on the fair values of the consideration given.

The financial statements of the Company as of and for the year ended December 31, 2016, were approved by the Company’s Board of Directors on February 27, 2017.

1) Amendments to K-IFRS and new interpretations that are mandatorily effective for the current year

Amendments to K-IFRS 1110 – Consolidated Financial Statements & K-IFRS 1112 Disclosure of interests in other entities & K-IFRS 1028 Investment in associates

The amendments clarify that in applying the equity method of accounting to an associate or a joint venture that is an investment entity, an investor may retain the fair value measurements that the associate or joint venture used for its subsidiaries. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company’s financial statements.

Amendments to K-IFRS 1111 – Accounting for Acquisitions of Interests in Joint Operations

The amendments to K-IFRS 1111 provide guidance on how to account for the acquisition of a joint operation that constitutes a business as defined in K-IFRS 1103 Business Combinations. A joint operator is also required to disclose the relevant information required by K-IFRS 1103 and other standards for business combinations. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company’s financial statements.

Amendments to K-IFRS 1001 – Disclosure Initiative

The amendments to K-IFRS 1001 clarify the concept of applying materiality in practice and restrict an entity reducing the understandability of its financial statements by obscuring material information with immaterial information or by aggregating material items that have different natures or functions. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company’s financial statements.

Amendments to K-IFRS 1016 – Property, Plant and Equipment

The amendments to K-IFRS 1016 prohibit the Company from using a revenue-based depreciation method for items of property, plant and equipment. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company’s financial statements.

Amendments to K-IFRS 1038 – Intangible Assets

The amendments to K-IFRS 1038 do not allow presumption that revenue is an appropriate basis for the amortization of intangible assets, which the presumption can only be limited when the intangible asset is expressed as a measure of revenue or when it can be demonstrated that revenue and consumption of the economic benefits of the intangible asset are highly correlated. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company’s financial statements.

Amendments to K-IFRS 1016 – Property, plant and equipment & K-IFRS 1041 Agriculture: Bearer Plants

The amendments to K-IFRS 1016 '*Property, Plant and Equipment*' and K-IFRS 1041 '*Agriculture*' define a bearer plant and require biological assets that meet the definition of a bearer plant to be accounted for as property, plant and equipment in accordance with K-IFRS 1016, instead of K-IFRS 1041. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company's financial statements.

Annual Improvements to K-IFRS 2012-2014 Cycle

The annual improvements include amendments to a number of K-IFRSs. The amendments introduce specific guidance in K-IFRS 1105 Non-current Assets Held for Sale and Discontinued Operations when an entity reclassifies an asset (or disposal group) from held for sale to held for distribution to owners (or vice versa); such a change is considered as a continuation of the original plan of disposal, and not as a change to a plan of sale. Other amendments in the annual improvements include K-IFRS 1107 Financial Instruments: Disclosures, K-IFRS 1019 Employee Benefits, and K-IFRS 1034 Interim Financial Reporting. The application of these amendments has no material impact on the disclosures or the amounts recognized in the Company's financial statements.

2) New and revised K-IFRSs in issue but not yet effective

The Company has not applied the following new and revised K-IFRSs that have been issued but are not yet effective.

Amendments to K-IFRS 1109 – Financial Instruments

The amendments to K-IFRS 1109 contain the requirements for the classification and measurement of financial assets and financial liabilities based on a business model whose objective is achieved both by collecting contractual cash flows and selling financial assets and based on the contractual terms that give rise on specified dates to cash flows, impairment methodology based on the expected credit losses, and broadened types of instruments that qualify as hedging instruments and the types of risk components of non-financial items that are eligible for hedge accounting and the change of the hedge effectiveness test. The amendments are effective for annual periods beginning on or after January 1, 2018.

Amendments to K-IFRS 1115 – Revenue from Contracts with Customers

The core principle under K-IFRS 1115 is that an entity should recognize revenue to depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity expects to be entitled in exchange for those goods or services. The amendments introduces a 5-step approach to revenue recognition and measurement: 1) Identify the contract with a customer, 2) Identify the performance obligations in the contract, 3) Determine the transaction price, 4) Allocate the transaction price to the performance obligations in the contract and, 5) Recognize revenue when (or as) the entity satisfies a performance obligation. This standard will supersede K-IFRS 1011 - Construction Contracts, K-IFRS 1018- Revenue, K-IFRS 2113 - Customer Loyalty Programmes, K-IFRS 2115-Agreements for the Construction of Real Estate, K-IFRS 2118 - Transfers of Assets from Customers, and K-IFRS 2031-Revenue-Barter Transactions Involving Advertising Services. The amendments are effective for annual periods beginning on or after January 1, 2018.

Amendments to K-IFRS 1102 – Share-based Payment

The amendments include: 1) when measuring the fair value of share-based payment, the effects of vesting and non-vesting conditions on the measurement of cash-settled share-based payment should be consistent with the measurement of equity-settled share-based payment, 2) Share-based payment transaction in which the Company settles the share-based payment arrangement net by withholding a specified portion of the equity instruments per statutory tax withholding requirements would be classified as equity-settled in its entirety, if otherwise would be classified as equity-settled without the net settlement feature, and 3) when a cash-settled share-based payment changes to an equity-settled share-based payment because of modifications of the terms and conditions, the original liability recognized is derecognized and the equity-settled share-based payment is recognized at the modification date fair value. Any difference between the carrying amount of the liability at the modification date and the amount recognized in equity at the same date would be recognized in profit and loss immediately. The amendments are effective for annual periods beginning on or after January 1, 2018.

Amendments to K-IFRS 1007 – Statement of Cash Flows

The amendments require that changes in liabilities arising from financial activities are disclosed. The amendments are effective for annual periods beginning on or after January 1, 2017.

Amendments to K-IFRS 1012 – Income Taxes

The amendments clarify that unrealized losses on fixed-rate debt instruments measured at fair value and measured at cost for tax purposes give rise to a deductible temporary difference regardless of whether the holder expects to recover the carrying amount of the debt instrument by sale or by use and that the estimate of probable future taxable profit may include the recovery of some of the assets for more than their carrying amount. When the Company assesses whether there will be sufficient taxable profit, the Company should compare the deductible temporary differences with future taxable profit that excludes tax deductions resulting from the reversal of those deductible temporary differences. The amendments are effective for annual periods beginning on or after January 1, 2017.

The Company is currently evaluating the impact of above-mentioned enactments and amendments on the Company's financial statements.

(2) Investments in associates and joint ventures

An associate is an entity over which the Company has significant influence. Significant influence is the power to participate in the financial and operating policy decisions of the investee but is not control or joint control over those policies.

A joint venture is a joint arrangement whereby the parties that have joint control of the arrangement have rights to the net assets of the joint arrangement. Joint control is the contractually agreed sharing of control of an arrangement, which exists only when decisions about the relevant activities require unanimous consent of the parties sharing control.

The results and assets and liabilities of associates or joint ventures are incorporated in these financial statements using the equity-accounted investees, except when the investment is classified as held for sale, in which case it is accounted for in accordance with K-IFRS 1105, *Non-current Assets Held for Sale and Discontinued Operations*. Under the equity method, an investment in an associate or a joint venture is initially recognized in the statement of financial position at cost and adjusted thereafter to recognize the Company's share of the profit or loss and other comprehensive income of the associate or joint venture. When the Company's share of losses of an associate or a joint venture exceeds the Company's interest in that associate or joint venture (which includes any long-term interests that, in substance, form part of the Company's net investment in the associate or joint venture), the Company discontinues recognizing its share of further losses. Additional losses are recognized only to the extent that the Company has incurred legal or constructive obligations or made payments on behalf of the associate or joint venture.

Any excess of the cost of acquisition over the Company's share of the net fair value of the identifiable assets, liabilities and contingent liabilities of an associate or a joint venture recognized at the date of acquisition is recognized as goodwill, which is included within the carrying amount of the investment. Any excess of the Company's share of the net fair value of the identifiable assets, liabilities and contingent liabilities over the cost of acquisition, after reassessment, is recognized immediately in profit or loss.

Upon disposal of an associate or a joint venture that results in the Company losing significant influence over that associate or joint venture, any retained investment is measured at fair value at that date and the fair value is regarded as its fair value on initial recognition as a financial asset in accordance with K-IFRS 1039 *Financial Instruments: Recognition and Measurement*. The difference between the previous carrying amount of the associate or joint venture attributable to the retained interest and its fair value is included in the determination of the gain or loss on disposal of the associate or joint venture. In addition, the Company accounts for all amounts previously recognized in other comprehensive income in relation to that associate or joint venture on the same basis we would be required if that associate or joint venture had directly disposed of the related assets or liabilities. Therefore, if a gain or loss previously recognized in other comprehensive income by that associate or joint venture would be reclassified to profit or loss on the disposal of the related assets or liabilities, the Company reclassifies the gain or loss from equity to profit or loss (as reclassification adjustment) when it loses significant influence over that associate or joint venture.

When the Company reduces its ownership interest in an associate or a joint venture but the Company continues to use the equity method, the Company reclassifies the profit or loss the proportion of the gain or loss that had previously been recognized in other comprehensive income relating to that reduction in ownership interest if that gain or loss would be reclassified to profit or loss on the disposal of the related assets or liabilities. In addition, the Company applies K-IFRS 1105, *Non-current Assets Held for Sale and Discontinued Operations*, to a portion of investment in an associate or a joint venture that meets the criteria to be classified as held for sale.

The requirements of K-IFRS 1039, *Financial Instruments: Recognition and Measurement*, are applied to determine whether it is necessary to recognize any impairment loss with respect to the Company's investment in an associate or a joint venture. When necessary, the entire carrying amount of the investment (including goodwill) is tested for impairment in accordance with K-IFRS 1036, *Impairment of Assets*, by comparing its recoverable amount (higher of value in use and fair value less costs to sell) with its carrying amount, any impairment loss recognized forms part of the carrying amount of the investment. Any reversal of that impairment loss is recognized in accordance with K-IFRS 1036, *Impairment of Assets*, to the extent that the recoverable amount of the investment subsequently increases.

The Company continues to use the equity method when an investment in an associate becomes an investment in a joint venture or an investment in a joint venture becomes an investment in an associate. There is no remeasurement to fair value upon such changes in ownership interests.

When the Company transacts with an associate or a joint venture, profits and losses resulting from the transactions with the associate or joint venture are recognized in the Company's financial statements only to the extent of interests in the associate or joint venture that are not related to the Company.

(3) Interests in joint operations

A joint operation is a joint arrangement whereby the parties that have joint control of the arrangement have rights to the assets, and obligations for the liabilities, relating to the arrangement. Joint control is the contractually agreed sharing of control of an arrangement, which exists only when decisions about the relevant activities require unanimous consent of the parties sharing control.

When the Company undertakes its activities under joint operations, the Company, as a joint operator, recognizes in relation to its interest in a joint operation:

- its assets, including its share of any assets held jointly
- its liabilities, including its share of any liabilities incurred jointly
- its revenue from the sale of its share of the output arising from the joint operation
- its share of the revenue from the sale of the output by the joint operation
- its expenses, including its share of any expenses incurred jointly

The Company accounts for the assets, liabilities, revenues and expenses relating to its interest in a joint operation in accordance with the K-IFRSs applicable to the particular assets, liabilities, revenues and expenses.

When the Company transacts with a joint operation in which the Company is a joint operator (such as a sale or contribution of assets), the Company is considered to be conducting the transaction with the other parties to the joint operation, and gains and losses resulting from the transactions are recognized in the Company's financial statements only to the extent of other parties' interests in the joint operation.

When the Company transacts with a joint operation in which the Company is a joint operator (such as a purchase of assets), the Company does not recognize its share of the gains and losses until it resells those assets to a third party.

(4) Revenue recognition

Revenue is measured at the fair value of the consideration received or receivable. Revenue is reduced for estimated customer returns, rebates and other similar allowances. The Company recognizes revenue when the amount of revenue can be reliably measured; when it is probable that future economic benefits will flow to the entity; and when specific criteria have been met for each of the Company's activities.

Sales of goods and rendering services are separately recognized in revenue with distinguishable unit when more than two sales of goods and rendering service are made by one contract. The consideration of contract is separately allocated to each unit based on its fair value.

(5) Foreign currency translations

The non-consolidated financial statements of the Company are presented in the currency of the primary economic environment in which the entity operates (its functional currency). For the purpose of the financial statements, the results and financial position of each group entity are expressed in Korean won, which is the functional currency of the entity and the presentation currency for the non-consolidated financial statements.

In preparing the financial statements of the individual entities, transactions in currencies other than the entity's functional currency (foreign currency translations) are recognized at the rates of exchange prevailing at the dates of the transactions. At the end of each reporting period, monetary items denominated in foreign currency translations are retranslated at the rates prevailing at that date. Non-monetary items carried at fair value that are denominated in foreign currency translations are retranslated at the rates prevailing at the date when the fair value was determined. Non-monetary items that are measured in terms of historical cost in a foreign currency are not retranslated.

Exchange differences are recognized in profit or loss in the period in which they arise except for:

- exchange differences on foreign currency borrowings relating to assets under construction for future productive use, which are included in the cost of those assets when they are regarded as an adjustment to interest costs on those foreign currency borrowings
- exchange differences on transactions entered into in order to hedge certain foreign currency
- exchange differences on monetary items receivable from or payable to a foreign operation for which settlement is neither planned nor likely to occur (therefore forming part of the net investment in the foreign operation), which are recognized initially in other comprehensive income and reclassified from equity to profit or loss on disposal or partial disposal of the net investment

For the purpose of presenting financial statements, the assets and liabilities of the Company's foreign operations are expressed in Korean won using exchange rates prevailing at the end of the reporting period. Income and expense items are translated at the average exchange rates for the period, unless exchange rates fluctuated significantly during that period, in which case the exchange rates at the dates of the transactions are used. Exchange differences arising, if any, are recognized in other comprehensive income and accumulated in equity (attributed to non-controlling interests as appropriate).

On the disposal of a foreign operation (i.e., a disposal of the Company's entire interest in a foreign operation, or a disposal involving loss of control over a subsidiary that includes a foreign operation, or partial disposal of an interest in a joint arrangement or an associate that includes a foreign operation of which the retained interest becomes a financial asset), all of the accumulated exchange differences in respect of that operation attributable to the owners of the Company are reclassified to profit or loss. Any exchange differences that have previously been attributed to non-controlling interests are derecognized, but they are not reclassified to profit or loss.

In the case of a partial disposal (i.e., no loss of control) of a subsidiary that includes a foreign operation, the proportionate share of accumulated exchange differences are reattributed to non-controlling interests in equity and are not recognized in profit or loss. For all other partial disposals (i.e., partial disposals of associates or joint arrangements that do not result in the Company losing significant influence or joint control), the proportionate share of the accumulated exchange differences is reclassified to profit or loss.

Goodwill and fair value adjustments arising on the acquisition of a foreign operation are treated as assets and liabilities of the foreign operation and translated at the closing rate. Exchange differences arising are recognized in equity.

(6) Borrowing costs

Borrowing costs directly attributable to the acquisition, construction or production of qualifying assets, which are assets that necessarily take a substantial period of time to get ready for their intended use or sale, are added to the cost of those assets, until such time as the assets are substantially ready for their intended use or sale.

Investment income earned on the temporary investment of specific borrowings pending their expenditure on qualifying assets is deducted from the borrowing costs eligible for capitalization.

All other borrowing costs are recognized in profit or loss in the period in which they are incurred.

(7) Government grants

Government grants are not recognized until there is reasonable assurance that the Company will comply with the conditions attached to them and that the grants will be received.

The benefit of a government loan at a below-market rate of interest is treated as a government grant, measured as the difference between proceeds received and the fair value of the loan based on prevailing market interest rates.

Government grants related to assets are presented in the statement of financial position by deducting the grant from the carrying amount of the asset. The related grant is recognized in profit or loss over the life of a depreciable asset as a reduced depreciation expense.

Government grants related to income are recognized in profit or loss on a systematic basis over the periods in which the Company recognizes as expenses the related costs for which the grants are intended to compensate. Government grants that are receivable as compensation for expenses or losses already incurred or for the purpose of giving immediate financial support to the Company with no future related costs are recognized in profit or loss in the period in which they become receivable.

(8) Retirement benefit costs and termination benefits

Contributions to defined contribution retirement benefit plans are recognized as an expense when employees have rendered service entitling them to the contributions.

For defined benefit retirement benefit plans, the cost of providing benefits is determined using the Projected Unit Credit Method, with actuarial valuations being carried out at the end of each reporting period. Remeasurement, comprising actuarial gains and losses, the effect of the changes to the asset ceiling (if applicable) and the return on plan assets (excluding interest), is reflected immediately in the statement of financial position with a charge or credit recognized in other comprehensive income in the period in which they occur. Remeasurement recognized in other comprehensive income is reflected immediately in retained earnings and will not be reclassified to profit or loss. Past service cost is recognized in profit or loss in the period of a plan amendment. Net interest is calculated by applying the discount rate at the beginning of the period to the net defined benefit liability or asset. Defined benefit costs are composed of service cost (including current service cost, past service cost, as well as gains and losses on curtailments and settlements), net interest expenses (income), and remeasurement.

The Company presents the service cost and net interest expenses (income) components in profit or loss, and the remeasurement component in other comprehensive income. Curtailment gains and losses are accounted for as past service costs.

The retirement benefit obligation recognized in the statement of financial position represents the actual deficit or surplus in the Company's defined benefit plans. Any surplus resulting from this calculation is limited to the present value of any economic benefits available in the form of refunds from the plans or reductions in future contributions to the plans.

A liability for a termination benefit is recognized at the earlier of when the entity can no longer withdraw the offer of the termination benefit and when the entity recognizes any related restructuring costs.

(9) Taxation

Income tax expense represents the sum of the tax currently payable and deferred tax.

1) Current tax

The tax currently payable is based on taxable profit for the year. Taxable profit differs from profit as reported in the statement of profit or loss and comprehensive income because of items of income or expense that are taxable or deductible in other years and items that are never taxable or deductible. The Company's liability for current tax is calculated using tax rates that have been enacted or substantively enacted by the end of the reporting period.

2) Deferred tax

Deferred tax is recognized on temporary differences between the carrying amounts of assets and liabilities in the financial statements and the corresponding tax bases used in the computation of taxable profit. Deferred tax liabilities are generally recognized for all taxable temporary differences. Deferred tax assets are generally recognized for all deductible temporary differences to the extent that it is probable that taxable profits will be available against which those deductible temporary differences can be utilized. Such deferred tax assets and liabilities are not recognized if the temporary difference arises from goodwill or from the initial recognition (other than in a business combination) of other assets and liabilities in a transaction that affects neither the taxable profit nor the accounting profit.

Deferred tax liabilities are recognized for taxable temporary differences associated with investments in subsidiaries and associates, and interests in joint ventures, except where the Company is able to control the reversal of the temporary difference and it is probable that the temporary difference will not reverse in the foreseeable future. Deferred tax assets arising from deductible temporary differences associated with such investments and interests are only recognized to the extent that it is probable that there will be sufficient taxable profits against which to utilize the benefits of the temporary differences and they are expected to reverse in the foreseeable future.

The carrying amount of deferred tax assets is reviewed at the end of each reporting period and reduced to the extent that it is no longer probable that sufficient taxable profits will be available to allow all or part of the asset to be recovered.

Deferred tax assets and liabilities are measured at the tax rates that are expected to apply in the period in which the liability is settled or the asset realized, based on tax rates (and tax laws) that have been enacted or substantively enacted by the end of the reporting period. The measurement of deferred tax liabilities and assets reflects the tax consequences that would follow from the manner in which the Company expects, at the end of the reporting period, to recover or settle the carrying amount of its assets and liabilities.

Deferred tax assets and liabilities are offset if, and only if, the Company has a legally enforceable right to set off current tax assets against current tax liabilities, and the deferred tax assets and liabilities relate to income taxes levied by the same taxation authority on either the same taxable entity or different taxable entities which intend either to settle current tax liabilities and assets on a net basis, or to realize the assets and settle the liabilities simultaneously, in each future period in which significant amounts of deferred tax liabilities or assets are expected to be settled or recovered.

For the purpose of measuring deferred tax liabilities and deferred tax assets for investment properties that are measured using the fair value model, the carrying amounts of such properties are presumed to be recovered entirely through sale, unless the presumption is rebutted. The presumption is rebutted when the investment property is depreciable and is held within a business model whose objective is to consume substantially all of the economic benefits embodied in the investment properties over time, rather than through sale.

3) Current and deferred tax for the year

Current and deferred tax are recognized in profit or loss, except when they relate to items that are recognized in other comprehensive income or directly in equity, in which case, the current and deferred tax are also recognized in other comprehensive income or directly in equity respectively. Where current tax or deferred tax arises from the initial accounting for a business combination, the tax effect is included in the accounting for the business combination.

(10) Property, plant and equipment

Property, plant and equipment are stated at cost less subsequent accumulated depreciation and accumulated impairment losses. The cost of an item of property, plant and equipment is directly attributable to their purchase or construction, which includes any costs directly attributable to bringing the asset to the location and condition necessary for it to be capable of operating in the manner intended by management. It also includes the initial estimate of the costs of dismantling and removing the item and restoring the site on which it is located.

Subsequent costs are recognized in carrying amount of an asset or as a separate asset if it is probable that future economic benefits associated with the assets will flow into the Company and the cost of an asset can be measured reliably. Routine maintenance and repairs are expensed as incurred.

The Company does not depreciate land. Depreciation expense is computed using the straight-line method based on the estimated useful lives of the assets as follows:

	<u>Useful lives (years)</u>
Buildings and structures	40
Machinery and production facilities	5–15
Vehicles	4
Other property, plant and equipment (“Other PP&E”)	4

If each part of an item of property, plant and equipment has a cost that is significant in relation to the total cost of the item, it is depreciated separately.

The Company reviews the depreciation method, the estimated useful lives and residual values of property, plant and equipment at the end of each annual reporting period. If expectations differ from previous estimates, the changes are accounted for as a change in an accounting estimate.

An item of property, plant and equipment is derecognized upon disposal or when no future economic benefits are expected to arise from the continued use of the asset. Any gain or loss arising on derecognition of the property (calculated as the difference between the net disposal proceeds and the carrying amount of the asset) is included in profit or loss in the period in which the property is derecognized.

(11) Intangible assets

Intangible assets are initially recognized at acquisition cost and presented in net of amortization and impairment loss. An asset is amortized using a straight-line method over useful lives from the point the capitalized asset is ready for use. Meanwhile, some intangible asset was applied indefinite useful live which has unpredictable period.

	<u>Useful lives (years)</u>
Development costs	10
Software	5
Other intangible assets	5
Land use rights	Indefinite

1) Intangible assets acquired separately

Intangible assets with finite useful lives that are acquired separately are carried at cost, less accumulated amortization and accumulated impairment losses. Amortization is recognized on a straight-line basis over their estimated useful lives. The estimated useful life and amortization method are reviewed at the end of each reporting period, with the effect of any changes in estimate being accounted for on a prospective basis. Intangible assets with indefinite useful lives that are acquired separately are carried at cost, less accumulated impairment losses.

2) Internally generated intangible assets - research and development expenditure

Expenditure on research activities is recognized as an expense in the period in which it is incurred.

Expenditure arising from development (or from the development phase of an internal project) is recognized as an intangible asset if, only if, the development project is designed to produce new or substantially improved products, and the Company can demonstrate the technical and economical feasibility and measure reliably the resources attributable to the intangible asset during its development.

The amount initially recognized for internally generated intangible assets is the sum of the expenditure incurred from the date when the intangible asset first meets the recognition criteria. Where no internally generated intangible asset can be recognized, development expenditure is recognized in profit or loss in the period in which it is incurred.

Subsequent to initial recognition, internally generated intangible assets are reported at cost, less accumulated amortization and accumulated impairment losses, on the same basis as intangible assets that are acquired separately.

3) Intangible assets acquired in a business combination

Intangible assets that are acquired in a business combination are recognized separately from goodwill and are initially recognized at their fair value at the acquisition date (which is regarded as their cost). Subsequent to initial recognition, intangible assets acquired in a business combination are reported at cost, less accumulated amortization and accumulated impairment losses, on the same basis as intangible assets that are acquired separately.

4) Derecognition of intangible assets

An intangible asset is derecognized on disposal, or when no future economic benefits are expected from its use. Gains or losses arising from derecognition of an intangible asset, measured as the difference between the net disposal proceeds and the carrying amount of the asset, and are recognized in profit or loss when the asset is derecognized.

(12) Impairment of tangible and intangible assets other than goodwill

At the end of each reporting period, the Company reviews the carrying amounts of its tangible and intangible assets to determine whether there is any indication that those assets have suffered an impairment loss. If any such indication exists, the recoverable amount of the asset is estimated in order to determine the extent of the impairment loss (if any). When it is not possible to estimate the recoverable amount of an individual asset, the Company estimates the recoverable amount of the cash generating unit to which the asset belongs. Where a reasonable and consistent basis of allocation can be identified, corporate assets are also allocated to individual cash-generating units ("CGU"), or otherwise they are allocated to the smallest group of CGU for which a reasonable and consistent allocation basis can be identified.

Intangible assets with indefinite useful lives and intangible assets not yet available for use are tested for impairment at least annually, and whenever there is an indication that the asset may be impaired.

Recoverable amount is the higher of fair value, less costs to sell and value in use. If the recoverable amount of an asset (or a CGU) is estimated to be less than its carrying amount, the carrying amount of the asset (or the CGU) is reduced to its recoverable amount and the reduced amount is recognized in profit or loss.

Where an impairment loss subsequently reverses, the carrying amount of the asset (or a CGU) is increased to the revised estimate of its recoverable amount, but so that the increased carrying amount does not exceed the carrying amount that would have been determined had no impairment loss been recognized for the asset (or the CGU) in prior years. A reversal of an impairment loss is recognized immediately in profit or loss.

(13) Inventories

Inventories are stated at the lower of cost and net realizable value. Cost of inventories, except for those in transit, is measured under the weighted-average method on a first-in, first-out basis and consists of the purchase price, cost of conversion and other costs incurred in bringing the inventories to their present location and condition. Net realizable value represents the estimated selling price for inventories less all estimated costs of completion and costs necessary to make the sale.

When inventories are sold, the carrying amount of those inventories is recognized as an expense (cost of sales) in the period in which the related revenue is recognized. The amount of any write-down of inventories to net realizable value and all losses of inventories is recognized as an expense in the period the write-down or loss occurs. The amount of any reversal of any write-down of inventories, arising from an increase in net realizable value, is recognized as a reduction in the amount of inventories recognized as an expense in the period in which the reversal occurs.

(14) Provisions

Provisions are recognized when the Company has a present obligation (legal or constructive) as a result of a past event and it is probable that the Company will be required to settle the obligation, and a reliable estimate can be made of the amount of the obligation.

The amount recognized as a provision is the best estimate of the consideration required to settle the present obligation at the end of the reporting period, taking into account the risks and uncertainties surrounding the obligation. When a provision is measured using the cash flows estimated to settle the present obligation, its carrying amount is the present value of those cash flows (where the effect of the time value of money is material). The discount rate used is a pretax rate that reflects current market assessments of the time value of money and the risks specific to the liability. Where discounting is used, the increase in the provision due to the passage is recognized in profit or loss as borrowing cost.

When some or all of the economic benefits required to settle a provision are expected to be recovered from a third party, a receivable is recognized as an asset if it is virtually certain that reimbursement will be received and the amount of the receivable can be measured reliably.

At the end of each reporting period, the remaining provision balance is reviewed and assessed to determine if the current best estimate is being recognized. If the existence of an obligation to transfer economic benefit is no longer probable, the related provision is reversed during the period.

(15) Financial Instruments

Financial assets and financial liabilities are recognized when an entity becomes a party to the contractual provisions of the instruments. Financial assets and financial liabilities are initially measured at fair value. Transaction costs that are directly attributable to the acquisition or issue of financial assets and financial liabilities are added to or deducted from the fair value of the financial assets or financial liabilities, as appropriate, on initial recognition. Transaction costs directly attributable to the acquisition of financial assets or financial liabilities at fair value through profit or loss ("FVTPL") are recognized immediately in profit or loss.

All regular-way purchases or sales of financial assets are recognized and derecognized on a trade-date basis. Regular-way purchases or sales are purchases or sales of financial assets that require delivery of assets within the time frame established by regulation or convention in the marketplace.

Financial assets are classified into the following specified categories: 'financial assets at FVTPL,' 'held-to-maturity investments,' 'available-for-sale ("AFS") financial assets' and 'loans and receivables.' The classification depends on the nature and purpose of the financial assets and is determined at the time of initial recognition.

1) Effective interest method

The effective interest method is a method of calculating the amortized cost of a debt instrument and of allocating interest income over the relevant period. The effective interest rate is the rate that exactly discounts estimated future cash receipts (including all fees and points paid or received that form an integral part of the effective interest rate, transaction costs and other premiums or discounts) through the expected life of the debt instrument, or, where appropriate, a shorter period, to the net carrying amount on initial recognition.

Income is recognized on an effective interest basis for debt instruments other than those financial assets classified as at FVTPL.

2) Financial assets at FVTPL

Financial assets are classified as at FVTPL when the financial asset is contingent consideration that may be paid by an acquirer as part of business combination to which K-IFRS 1103 applies,

A financial asset is classified as held for trading if:

- it has been acquired principally for the purpose of selling it in the near term
- on initial recognition it is part of a portfolio of identified financial instruments that the Company manages together and has a recent actual pattern of short-term profit-taking
- it is a derivative that is not designated and effective as a hedging instrument

A financial asset other than a financial asset held for trading or contingent consideration that may be paid by an acquirer as part of a business combination may be designated as at FVTPL upon initial recognition if:

- such designation eliminates or significantly reduces a measurement or recognition inconsistency that would otherwise arise
- the financial asset forms part of a group of financial assets or financial liabilities or both, which is managed and its performance is evaluated on a fair value basis, in accordance with the Company's documented risk management or investment strategy, and information about the grouping is provided internally on that basis
- it forms part of a contract containing one or more embedded derivatives, and K-IFRS 1039 Financial Instruments: Recognition and Measurement permits the entire combined contract (asset or liability) to be designated as at FVTPL

Financial assets at FVTPL are stated at fair value, with any gains or losses arising on remeasurement recognized in profit or loss. The net gain or loss recognized in profit or loss incorporates any dividend or interest earned on the financial asset and is included in the 'other gains and losses' line item in the statements of comprehensive income.

3) Held-to-maturity investments

Non-derivative financial assets with fixed or determinable payments and fixed-maturity dates that the Company has the positive intent and ability to hold to maturity are classified as held-to-maturity investments. Held-to-maturity investments are measured at amortized cost using the effective interest method less any impairment, with revenue recognized on an effective yield basis.

4) AFS financial assets

AFS financial assets are non-derivatives that are either designated as AFS or are not classified as (a) loans and receivables, (b) held-to-maturity investments or (c) financial assets at FVTPL.

They are subsequently measured at fair value at the end of each reporting period. Changes in the carrying amount of AFS monetary financial assets relating to changes in foreign currency rates (see below), interest income calculated using the effective interest method and dividends on AFS equity investments are recognized in profit or loss. Other changes in the carrying amount of AFS financial assets are recognized in other comprehensive income (as [investments revaluation reserve]). When the investment is disposed of or is determined to be impaired, the cumulative gain or loss previously accumulated in other comprehensive income is reclassified to profit or loss.

Dividends on AFS equity instruments are recognized in profit or loss when the Company's right to receive the dividends is established.

The fair value of AFS monetary financial assets denominated in a foreign currency is determined in that foreign currency and translated at the spot rate prevailing at the end of the reporting period. The foreign exchange gains and losses that are recognized in profit or loss are determined based on the amortized cost of the monetary asset. Other foreign exchange gains and losses are recognized in other comprehensive income.

AFS equity investments that do not have a quoted market price in an active market and whose fair value cannot be reliably measured and derivatives that are linked to and must be settled by delivery of such unquoted equity investments are measured at cost, less any identified impairment losses at the end of each reporting period.

5) Loans and receivables

Trade receivables, loans and other receivables that have fixed or determinable payments that are not quoted in an active market are classified as 'loans and receivables.' Loans and receivables are measured at amortized cost using the effective interest method, less any impairment. Interest income is recognized by applying the effective interest rate, except for short-term receivables when the effect of discounting is immaterial.

6) Impairment of financial assets

Financial assets, other than those at FVTPL, are assessed for indicators of impairment at the end of each reporting period. Financial assets are considered to be impaired when there is objective evidence that more events that occurred after the initial recognition of the financial asset, the estimated future cash flows of the investment have been affected.

For AFS equity investments, a significant or prolonged decline in the fair value of the security below its cost is considered to be objective evidence of impairment.

For all other financial assets, objective evidence of impairment includes:

- significant financial difficulty of the issuer or counterparty
- default or delinquency in interest or principal payments
- it becoming probable that the borrower will enter bankruptcy or financial reorganization
- the disappearance of an active market for that financial asset because of financial difficulties

For financial assets carried at amortized cost, the amount of the impairment loss recognized is the difference between the asset's carrying amount and the present value of estimated future cash flows, discounted at the financial asset's original effective interest rate.

For financial assets that are carried at cost, the amount of the impairment loss recognized is the difference between the asset's carrying amount and the present value of estimated future cash flows, discounted at the current market rate of return for a similar financial asset. Such impairment loss will not be reversed in subsequent periods.

The carrying amount of the financial asset is reduced by the impairment loss directly for all financial assets with the exception of trade receivables, where the carrying amount is reduced through the use of an allowance account. When a trade receivable is considered uncollectible, it is written off against the allowance account. Subsequent recoveries of amounts previously written off are credited against the allowance account. Changes in the carrying amount of the allowance account are recognized in profit or loss.

When an AFS financial asset is considered to be impaired, cumulative gains or losses previously recognized in other comprehensive income are reclassified to profit or loss in the period.

For financial assets measured at amortized cost, if, in a subsequent period, the amount of the impairment loss decreases and the decrease can be related objectively to an event occurring after the impairment was recognized, the previously recognized impairment loss is reversed through profit or loss to the extent that the carrying amount of the investment at the date the impairment is reversed does not exceed what the amortized cost would have been had the impairment not been recognized.

In respect of AFS equity securities, impairment losses previously recognized in profit or loss are not reversed through profit or loss. Any increase in fair value subsequent to an impairment loss is recognized in other comprehensive income. In respect of AFS debt securities, impairment losses are subsequently reversed through profit or loss if an increase in the fair value of the investment can be objectively related to an event occurring after the recognition of the impairment loss.

7) Derecognition of financial assets

The Company derecognizes a financial asset only when the contractual rights to the cash flows from the asset expire, or when it transfers the financial asset and substantially all the risks and rewards of ownership of the asset to another entity. If the Company neither transfers nor retains substantially all the risks and rewards of ownership and continues to control the transferred asset, the Company recognizes its retained interest in the asset and an associated liability for amounts it may have to pay. If the Company retains substantially all the risks and rewards of ownership of a transferred financial asset, the Company continues to recognize the financial asset and also recognizes a collateralized borrowing for the proceeds received.

On derecognition of a financial asset in its entirety, the difference between the asset's carrying amount and the sum of the consideration received and receivable and the cumulated gain or loss that had been recognized in other comprehensive income and accumulated in equity is recognized in profit or loss.

On derecognition of a financial asset other than in its entirety (e.g., when the Company retains an option to repurchase part of a transferred asset, or it retains a residual interest and such retained interest indicates that the transferor has neither transferred nor retained substantially all the risks and rewards of ownership and has retained control of the transferred asset), the Company allocates the previous carrying amount of the financial asset between the part it continues to recognize under continuing involvement and the part it no longer recognizes on the basis of the relative fair value of those parts on the date of the transfer. The difference between the carrying amount allocated to the part that is no longer recognized and the sum of the consideration received for the part that is no longer recognized and any cumulative gain or loss allocated to it that had been recognized in other comprehensive income is recognized in profit or loss. A cumulative gain or loss that had been recognized in other comprehensive income is allocated between the part that continues to be recognized and the part that is no longer recognized on the basis of the relative fair value of those parts.

(16) Financial liabilities and equity instruments

1) Classification as debt or equity

Debt and equity instruments are classified as either financial liabilities or as equity in accordance with the substance of the contractual arrangement and the definitions of financial liability and an equity instrument.

2) Equity instruments

An equity instrument is any contract that evidences a residual interest in the assets of an entity after deducting all of its liabilities. Equity instruments issued by the Company are recognized at the proceeds received, net of direct issue costs.

Repurchase of the Company's own equity instruments is recognized and deducted directly in equity. No gain or loss is recognized in profit or loss on the purchase, sale, issue or cancellation of the Company's own equity instruments.

3) Compound instruments

The component parts of compound instruments (convertible bonds) issued by the Company are classified separately as financial liabilities and equity in accordance with the substance of the contractual arrangements and the definitions of a financial liability and equity instrument. Conversion option that will be settled by the exchange of a fixed amount of cash or another financial asset for a fixed number of the Company's own equity instruments is an equity instrument.

At the date of issue, the fair value of the liability component is estimated using the prevailing market interest rate for a similar non-convertible instrument. This amount is recorded as a liability on an amortized cost basis using the effective interest method until extinguished upon conversion or at the instrument's maturity date.

The conversion option classified as equity is determined by deducting the amount of the liability component from the fair value of the compound instrument as a whole. This is recognized and included in equity, net of income tax effects, and is not subsequently remeasured. In addition, the conversion option classified as equity will remain in equity until the conversion option is exercised, in which case, the balance recognized in equity will be transferred to share premium. No gain or loss is recognized in profit or loss upon conversion or expiration of the conversion option.

Transaction costs that relate to the issue of the convertible notes are allocated to liability and equity components in proportion to the allocation of the gross proceeds. Transaction costs relating to equity component are recognized directly in equity. Transaction costs relating to the liability component are included in the carrying amount of the liability component and are amortized over the lives of the convertible notes using the effective interest method.

4) Financial liabilities

Financial liabilities are recognized when the Company becomes a party to the contractual provisions of the instruments. Financial liabilities are initially measured at fair value. Transaction cost that is directly attributable to the issue of financial liabilities is deducted from the fair value of the financial liabilities on initial recognition. Transaction cost directly attributable to acquisition of financial liabilities at FVTPL are recognized immediately in profit or loss.

Financial liabilities are classified as either financial liabilities at FVTPL or other financial liabilities.

5) Financial liabilities at FVTPL

Financial liabilities are classified as at FVTPL when the financial liability is contingent consideration that may be paid by an acquirer as part of a business combination to which K-IFRS 1103 applies, or held for trading, or it is designated as at FVTPL.

A financial liability is classified as held for trading if:

- it has been acquired principally for the purpose of repurchasing it in the near term
- on initial recognition it is part of a portfolio of identified financial instruments that the Company manages together and has a recent actual pattern of short-term profit-taking
- it is a derivative that is not designated and effective as a hedging instrument

A financial liability other than a financial liability held for trading or contingent consideration that may be paid by an acquirer as part of a business combination may be designated as at FVTPL upon initial recognition if:

- such designation eliminates or significantly reduces a measurement or recognition inconsistency that would otherwise arise
- the financial liability forms part of a group of financial assets or financial liabilities or both, which is managed and its performance is evaluated on a fair value basis, in accordance with the Company's documented risk management or investment strategy, and information about the grouping is provided internally on that basis
- it forms part of a contract containing one or more embedded derivatives, and K-IFRS 1039, Financial Instruments: Recognition and Measurement, permits the entire combined contract (asset or liability) to be designated as at FVTPL

Financial liabilities at FVTPL are stated at fair value, with any gains or losses arising on remeasurement recognized in profit or loss. The net gain or loss recognized in profit or loss incorporates any interest paid on the financial liability and is included in the 'other gains and losses' line item in the statements of comprehensive income.

6) Other financial liabilities

Other financial liabilities are subsequently measured at amortized cost using the effective interest method, with interest expenses recognized on an effective yield basis.

The effective interest method is a method of calculating the amortized cost of a financial liability and of allocating interest expenses over the relevant period. The effective interest rate is the rate that exactly discounts estimated future cash payments including all fees and points paid or received (that form an integral part of the effective interest rate) and transaction costs and other premiums or discounts through the expected life of the financial liability, or (where appropriate) a shorter period, to the net carrying amount on initial recognition.

7) Financial guarantee contracts

A financial guarantee contract is a contract that requires the issuer to make specified payments to reimburse the holder for a loss it incurs because a specified debtor fails to make payments when due in accordance with the terms of a debt instrument.

Financial guarantee contract liabilities are initially measured at their fair values and, if not designated as at FVTPL, are subsequently measured at the higher of:

- the amount of the obligation under the contract, as determined in accordance with K-IFRS 1037, *Provisions, Contingent Liabilities and Contingent Assets*
- the amount initially recognized, less cumulative amortization recognized in accordance with the K-IFRS 1018, *Revenue*

8) Derecognition of financial liabilities

The Company derecognizes financial liabilities when the Company's obligations are discharged, canceled or they expire. The difference between the carrying amount of the financial liability derecognized and the consideration paid and payable is recognized in profit or loss.

(17) Fair value

Fair value is the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date, regardless of whether that price is directly observable or estimated using another valuation technique. In estimating the fair value of an asset or a liability, the Company takes into account the characteristics of the asset or liability if market participants would take those characteristics into account when pricing the asset or liability at the measurement date. Fair value for measurement and/or disclosure purposes in these financial statements is determined on such a basis, except for share-based payment transactions that are within the scope of K-IFRS 1102, *Share-based payment*, leasing transactions that are within the scope of K-IFRS 1017, *Leases*, and measurements that have some similarities to fair value but are not fair value, such as net realisable value in K-IFRS 1002, *Inventories*, or value in use in K-IFRS 1036, *Impairment of Assets*.

In addition, for financial reporting purposes, fair value measurements are categorized into Levels 1, 2 or 3 based on the degree to which the inputs to the fair value measurements are observable and the significance of the inputs to the fair value measurement in its entirety, which are described as follows:

- Level 1 inputs are quoted prices (unadjusted) in active markets for identical assets or liabilities that the Company can access at the measurement date.
- Level 2 inputs are inputs, other than quoted prices included within Level 1, that are observable for the asset or liability, either directly or indirectly.
- Level 3 inputs are unobservable inputs for the asset or liability.

3. CRITICAL ACCOUNTING JUDGEMENTS AND KEY SOURCES OF ESTIMATION UNCERTAINTY:

In the application of the Company's accounting policies, which are described in Note 2, management is required to make judgments, estimates and assumptions about the carrying amounts of assets and liabilities that are not readily apparent from other sources. Actual results may differ from these estimates.

The estimates and underlying assumptions are reviewed on an ongoing basis. Revisions to accounting estimates are recognized in the period in which the estimate is revised if the revision affects only that period or in the period of the revision and future periods if the revision affects both current and future periods.

(1) Critical judgments in applying accounting policies

Information about critical judgments in applying accounting policies (Note 3(2)) that have the most significant effect on the amounts recognized in the financial statements is included in the following note:

Note 11 – Consolidation scope: whether the Company has de facto control over an investee

(2) Key sources of estimation uncertainty

Information about assumptions and estimation uncertainties that have a significant risk of resulting in a material adjustment within the next financial year are included in the following notes:

Note 5 – Derivatives liabilities: measurement of fair value

Note 13 – Impairment test: recoverability of assets

Note 17 – Measurement of defined benefit obligations: key actuarial assumptions

Note 25 – Recognition of deferred tax assets: availability of future taxable profit against which carryforward tax losses can be used

Note 27 – Provisions and contingencies: key assumptions about likelihood and magnitude of an outflow of resources

4. OPERATING SEGMENTS:

- (1) The Company has presented operating segment in the financial statements in accordance with K-IFRS 1108. The Company was the consolidated entity and had two reportable segments as 1) 'Manufacturing biological pharmaceutical's and 2) 'development and commercialization of biosimilar' as of December 31, 2015. The Company has one reportable segment, 'Manufacturing biological medicines,' as of December 31, 2016, after when changing consolidation scope.

	2016	
	Biological pharmaceutical manufacturing sector	
	(In thousands of Korean won)	
Operating income	₩	294,622,021
Operating expenses(*)		325,044,659
Operating profit (loss)	₩	(30,422,638)

- (*) The amount includes the cost of sales and selling, general and administrative expenses.

	2015			
	Biological pharmaceutical manufacturing	Development and commercialization of biosimilar	Total	After consolidation adjustments
	(In thousands of Korean won)			
Operating income	₩ 67,376,232	₩ 23,901,796	₩ 91,278,028	₩ 91,278,029
Operating expenses(*)	110,482,297	184,997,630	295,479,927	294,920,472
Operating profit (loss)	₩ (43,106,065)	₩ (161,095,834)	₩ (204,201,899)	₩ (203,642,443)

- (*) The amount includes the cost of sales and selling, general and administrative expenses.

- (*) As of the end of the previous year, the development and commercialization of biopharmaceuticals have been excluded from the consolidation so that the related asset and liability has not been presented.

- (2) Amount of revenue for customers who have over 10% of the Company's total revenue, including three companies, are ₩292,833 million as of December 31, 2016. (₩91,278 million, including two companies, as of December 31, 2015).

5. FINANCIAL INSTRUMENTS:

(1) Financial assets

Categories of financial assets as of December 31, 2016 and 2015, are as follows:

		December 31, 2016		
		Cash and cash equivalents	Loan and receivables	Total
		(In thousands of Korean won)		
Current assets:				
Cash and cash equivalents	₩	230,337,601	₩ -	₩ 230,337,601
Short-term financial instruments		-	970,000,000	970,000,000
Trade and other receivables		-	75,630,820	75,630,820
Deposits provided		-	16,289,300	16,289,300
		<u>230,337,601</u>	<u>1,061,920,120</u>	<u>1,292,257,721</u>
Non-current assets:				
Long-term financial instruments		-	8,000	8,000
Deposits provided		-	21,475,000	21,475,000
		-	21,483,000	21,483,000
	₩	<u>230,337,601</u>	₩ <u>1,083,403,120</u>	₩ <u>1,313,740,721</u>

		December 31, 2015		
		Cash and cash equivalents	Loan and receivables	Total
		(In thousands of Korean won)		
Current assets:				
Cash and cash equivalents	₩	34,510,062	₩ -	₩ 34,510,062
Trade and other receivables		-	37,009,243	37,009,243
Deposits provided		-	-	-
		<u>34,510,062</u>	<u>37,009,243</u>	<u>71,519,305</u>
Non-current assets:				
Long-term financial instruments		-	8,000	8,000
Deposits provided		-	24,980,000	24,980,000
		-	24,988,000	24,988,000
	₩	<u>34,510,062</u>	₩ <u>61,997,243</u>	₩ <u>96,507,305</u>

(2) Financial liabilities

Categories of financial liabilities as of December 31, 2016 and 2015, are as follows:

		December 31, 2016	December 31, 2015
		(In thousands of Korean won)	
Current liabilities:			
Bond Issues	₩	169,917,877	₩ -
Borrowings		-	10,109,490
Trade and other payables		138,056,375	56,193,891
Current portion of long-term borrowings		233,776,207	-
Financial derivative liabilities(*)		<u>1,874,470,943</u>	<u>1,820,439,050</u>
		<u>2,416,221,402</u>	<u>1,886,742,431</u>
Non-current liabilities:			
Bond Issues		50,000,000	50,000,000
Borrowings		389,363,194	631,792,000
Trade and other payables		2,300,380	1,909,574
		<u>441,663,574</u>	<u>683,701,574</u>
	₩	<u>2,857,884,976</u>	₩ <u>2,570,444,005</u>

(*) The Company recognized the fair value of the call option on Samsung Bioepis Co., Ltd.'s shares are held as financial derivative liabilities. The financial derivative liabilities will expire in 2018, and the option price is the sum of the principal amount and accumulated interest. The value of the financial derivative liabilities may be changed significantly depending on the market situation and the exercise of the option.

(3) Fair values

The fair value of financial instruments, together with the carrying amount as of December 31, 2016 and 2015, are as follows:

Financial assets measured at amortized cost (*)

	December 31, 2016		December 31, 2015	
	Carrying amount	Fair value	Carrying amount	Fair value
	(In thousands of Korean won)			
Cash and cash equivalents	₩ 230,337,601	₩ 230,337,601	₩ 34,510,062	₩ 34,510,062
Short-term financial instruments	970,000,000	970,000,000	-	-
Trade and other receivables:	75,630,820	75,630,820	37,009,243	37,009,243
- Trade receivables	66,461,789	66,461,789	34,958,506	34,958,506
- Other receivables	6,022,257	6,022,257	1,977,821	1,977,821
- Accrued revenues	3,146,774	3,146,774	72,916	72,916
Long-term financial instruments	8,000	8,000	8,000	8,000
Deposits provided	37,764,300	37,764,300	24,980,000	24,980,000
	<u>₩ 1,313,740,721</u>	<u>₩ 1,313,740,721</u>	<u>₩ 96,507,305</u>	<u>₩ 96,507,305</u>

Financial liabilities measured at fair value

	December 31, 2016		December 31, 2015	
	Carrying amount	Fair value	Carrying amount	Fair value
	(In thousands of Korean won)			
Derivatives liabilities	₩1,874,470,943	₩1,874,470,943	₩ 1,820,439,050	₩ 1,820,439,050

Financial liabilities measured at amortized cost (*)

	December 31, 2016		December 31, 2015	
	Carrying amount	Fair value	Carrying amount	Fair value
	(In thousands of Korean won)			
Bond Issues	₩ 219,917,877	₩ 219,917,877	₩ 50,000,000	₩ 50,000,000
Short-term borrowings	-	-	10,109,490	10,109,490
Current portion of long-term borrowings	233,776,207	233,776,207	-	-
Long-term borrowings	389,363,194	389,363,194	631,792,000	631,792,000
Trade and other payables:	140,356,755	140,356,755	58,103,465	58,103,465
- Trade payables	1,521,209	1,521,209	358,223	358,223
- Other payables	113,489,469	113,489,469	27,495,078	27,495,078
- Accrued expenses	25,346,077	25,346,077	30,250,164	30,250,164
	<u>₩ 983,414,033</u>	<u>₩ 983,414,033</u>	<u>₩ 750,004,955</u>	<u>₩ 750,004,955</u>

(*) Financial assets and liabilities measured at amortized cost are not subsequently measured at fair value and management determines that the carrying amounts of these financial assets and financial liabilities in the statement of financial position are similar to their fair values.

(4) Fair value hierarchy

The fair value hierarchy of financial instruments carried at fair value as of December 31, 2016 and 2015, are as follows:

	December 31, 2016			
	Level 1	Level 2	Level 3	Total
	(In thousands of Korean won)			
Financial derivative liabilities	₩ -	₩ -	₩ 1,874,470,943	₩ 1,874,470,943

	December 31, 2015			
	Level 1	Level 2	Level 3	Total
	(In thousands of Korean won)			
Financial derivative liabilities	₩ -	₩ -	₩ 1,820,439,050	₩ 1,820,439,050

(5) Valuation techniques and inputs

Valuation techniques and significant inputs used in measuring the fair value of financial instruments classified as Level 3 as of December 31, 2016 and 2015, are as follows:

<u>December 31, 2016</u>		<u>December 31, 2015</u>		Valuation techniques	Significant unobservable inputs
<u>Fair value</u>					
(In thousands of Korean won)					
Derivative liabilities	₩ 1,874,470,943	₩ 1,820,439,050		Binomial model	Fair value of underlying asset (Samsung Bioepis Co., Ltd) (Note 5(2))

Valuation techniques are based on Risk-adjusted NPV (net present value) for the above assets.

<u>Fair value</u>		<u>Valuation technique</u>
<u>December 31, 2016</u>	<u>December 31, 2015</u>	
<u>(In thousands of Korean won)</u>		
₩ 5,119,370,710	₩ 4,808,578,367	Risk-adjusted NPV

On the other hand, such fair value of stockholdings is significant when input variables are maintained stable. However, if any unforeseeable input variables reasonably fluctuate on the reporting date, the fair value of the derivative may be affected. The additional profit or loss effects resulting from such change are as follows:

<u>December 31, 2016</u>	<u>Increase</u>	<u>Decrease</u>
	<u>(In 100 millions of Korean won)</u>	
Revenue growth rate (0.5% change)	(945)	846
Risk-adjusted discount rate(1% change)	3,028	(3,819)

(6) Financial assets income and expenses by categories

Details of financial instruments income and expenses for the years ended December 31, 2016 and 2015, are as follows:

	<u>Interest income and expenses</u>		<u>Loss on evaluation of derivatives</u>		<u>Other income and expenses(*)</u>		<u>Net income (expenses)</u>	
	<u>2016</u>	<u>2015</u>	<u>2016</u>	<u>2015</u>	<u>2016</u>	<u>2015</u>	<u>2016</u>	<u>2015</u>
	<u>(In thousands of Korean won)</u>							
Cash and cash equivalents	₩ 1,396,566	₩ 1,182,311	₩ -	₩ -	₩ (2,002,552)	₩ (20,439,713)	₩ (605,986)	₩ (19,257,402)
Short-term financial instruments	2,243,084	-	-	-	-	-	2,243,084	-
Loans and receivables	-	-	-	-	3,176,260	(87,035,310)	3,176,260	(87,035,310)
Financial liabilities measured at amortized cost	(19,254,373)	(2,650,470)	-	-	(3,961,606)	95,030,158	(23,215,979)	92,379,688
Derivative liabilities	-	-	(54,031,894)	(1,820,439,050)	-	-	(54,031,894)	(1,820,439,050)
	<u>₩ (15,614,723)</u>	<u>₩ (1,468,159)</u>	<u>₩ (54,031,894)</u>	<u>₩ (1,820,439,050)</u>	<u>₩ (2,787,898)</u>	<u>₩ (12,444,865)</u>	<u>₩ (72,434,515)</u>	<u>₩ (1,834,352,074)</u>

(*) Other income and expenses include gains (losses) on foreign currency translations and gains (losses) on foreign currency transactions.

6. CASH AND CASH EQUIVALENTS:

Cash and cash equivalents as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In thousands of Korean won)	
Demand deposits	₩ 30,337,601	₩ 34,510,062
Other cash equivalents	<u>200,000,000</u>	<u>-</u>
	<u>₩ 230,337,601</u>	<u>₩ 34,510,062</u>

7. LIMITED USAGE OF FINANCIAL INSTRUMENTS:

Limited usage of financial instruments as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>	<u>Note</u>
	(In thousands of Korean won)		
Long-term financial instruments	₩ 8,000	₩ 8,000	Guarantee deposits for checking account

8. TRADE AND OTHER RECEIVABLES:

- (1) Trade and other receivables are presented with net value that is the value after deducting allowance for doubtful account. Details of trade and other receivables as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In thousands of Korean won)	
Trade receivable	₩ 66,461,789	₩ 34,958,506
Non-trade receivable	6,022,257	1,977,821
Accrued income	<u>3,146,774</u>	<u>72,916</u>
	<u>₩ 75,630,820</u>	<u>₩ 37,009,243</u>

Aging analysis of unimpaired trade and other receivables as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In thousands of Korean won)	
Less than one months (*)	₩ 18,339,121	₩ 35,103,311
More than one month, less than six months (*)	57,142,373	1,905,932
More than six months, less than 12 months (*)	138,608	-
More than 12 months (*)	<u>10,718</u>	<u>-</u>
	<u>₩ 75,630,820</u>	<u>₩ 37,009,243</u>

(*) It means that the period has passed from the day of bond issue.

The Company does not recognize an allowance for doubtful accounts for certain trade and other receivables, which are not at maturity, in consideration of historical rate of default.

9. INVENTORIES:

Details of inventories as of December 31, 2016 and 2015, are as follows:

		December 31, 2016		
		Acquisition cost	Provision for loss on valuation	Carrying amounts
		(In thousands of Korean won)		
Finished goods	₩	102,379,516	₩ (8,413,216)	₩ 93,966,300
Work in progress		2,772,749	(2,425,743)	347,006
Raw materials		35,630,578	-	35,630,578
Supplies		34,276,096	-	34,276,096
	₩	175,058,939	₩ (10,838,959)	₩ 164,219,980

		December 31, 2015		
		Acquisition cost	Provision for loss on valuation	Carrying amounts
		(In thousands of Korean won)		
Finished goods	₩	91,512,181	₩ (11,271,591)	₩ 80,240,590
Work in progress		3,605,892	(842,761)	2,763,131
Raw materials		15,328,189	-	15,328,189
Supplies		20,845,430	-	20,845,430
	₩	131,291,692	₩ (12,114,352)	₩ 119,177,340

Provision for loss on valuation of ₩1,275,392 thousand were reversed for the year ended December 31, 2016. For the year ended December 31, 2015, inventories of ₩5,391,777 thousand were written down to net realizable value. The write-downs are included in cost of sales.

10. OTHER FINANCIAL ASSETS AND OTHER ASSETS:

Details of other financial assets and other assets as of December 31, 2016 and 2015, are as follows:

	December 31, 2016		December 31, 2015	
	Current	Non-current	Current	Non-current
	(In thousands of Korean won)			
Other financial assets:				
Deposit	₩ 16,289,300	₩ 21,475,000	₩ -	₩ 24,980,000
Other assets:				
Prepaid income taxes	149,511	-	63,136	-
Advance payments	23,189	-	16,534	-
Prepaid expense	4,774,776	747,220	2,077,579	570,553
Subtotal	4,947,476	747,220	2,157,249	570,553
Total	₩ 21,236,776	₩ 22,222,220	₩ 2,157,249	₩ 25,550,553

11. INVESTMENTS IN ASSOCIATES AND JOINT VENTURES:

(1) Investments in associates and joint ventures as of December 31, 2016 and 2015, are as follows:

Investee	Major workplaces	Primary business	Fiscal date	December 31, 2016		December 31, 2015	
				Book value	Percentage of ownership	Book value	Percentage of ownership
(In thousands of Korean won)							
Archigen Biotech Limited	The United Kingdom	Bio-medicine business	2016.12.31	₩ 48,832,293	50.00%	₩ 28,961,024	50.00%
Samsung Bioepis Co. Ltd.(*)	The Republic of Korea	Biosimilar research and commercialization	2016.12.31	4,895,523,997	93.31%	4,808,578,367	91.20%
Total				₩ 4,944,356,290		₩ 4,837,539,391	

(*) Samsung Bioepis Co., Ltd. was set up with Biogen Inc. in 2012. Samsung Bioepis Co., Ltd. had been classified as a subsidiary until 2014. During 2015, as potential voting rights held by Biogen Inc. were changed to substantive rights to Samsung Bioepis Co., Ltd., the Company lost control over Samsung Bioepis Co., Ltd. If Biogen Therapeutics Inc. does not exercise options until the expiration (until the end of 2018), the judgment of control may be subject to loss and may result in loss (see Notes 1 and 5).

(2) Changes in investments in associates and joint ventures for the years ended December 31, 2016 and 2015, are as follows:

Investee	2016				
	Beginning balance	Acquisition	Share of loss of equity-accounted investees	Capital change	Ending balance
(In thousands of Korean won)					
Archigen Biotech Limited	₩ 28,961,024	₩ 34,653,000	₩ (17,477,034)	₩ 2,695,303	₩ 48,832,293
Samsung Bioepis Co., Ltd.(*)	4,808,578,367	200,000,000	(111,745,616)	(1,308,754)	4,895,523,997

(*) The Company's interest in Samsung Bioepis Co., Ltd. is recognized at fair value, which was measured based on DCF (Discounted Cash Flow).

Investee	2015				
	Beginning balance	Acquisition	Share of loss of equity-accounted investees	Capital change	Ending balance
(In thousands of Korean won)					
Archigen Biotech Limited	₩ 53,921,974	₩ -	₩ (19,395,735)	₩ (5,565,215)	₩ 28,961,024
Samsung Bioepis Co., Ltd.(*)	-	4,808,578,367	-	-	4,808,578,367

(*) The Company's interest in Samsung Bioepis Co., Ltd. is recognized at fair value, which was measured based on DCF (Discounted Cash Flow).

- (3) Financial information of investments in associates and joint ventures for the years ended December 31, 2016 and 2015, are as follows:

Investee	2016					
	Assets	Liabilities	Operating loss	Net loss	Other comprehensive loss	Total comprehensive loss
			(In thousands of Korean won)			
Archigen Biotech Limited	₩ 163,752,971	₩ 7,851,938	₩ -	₩ (34,939,862)	₩ -	₩ (34,939,862)
Samsung Bioepis Co., Ltd.(*)	1,096,519,052	713,879,598	147,476,750	(106,459,587)	(1,336,739)	(107,796,326)

- (*) This is the consolidated financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

Investee	2015					
	Assets	Liabilities	Operating profit (loss)	Net loss	Other comprehensive loss	Total comprehensive loss
			(In thousands of Korean won)			
Archigen Biotech Limited	₩ 119,418,165	₩ 3,259,668	₩ (28,725,844)	₩ (38,791,470)	₩ -	₩ (38,791,470)
Samsung Bioepis Co., Ltd.(*)	654,155,058	363,623,248	23,901,796	(166,637,271)	(2,508,135)	(169,145,406)

- (*) This is the consolidated financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

- (4) Reconciliation of the investees' financial information to the carrying value of the Company's interest in investments in associates and joint ventures as of December 31, 2016 and 2015, are as follows:

Investee	December 31, 2016					
	Net assets	Percentage of ownership	Company's share of net assets	Investment difference(*)	Elimination of unrealized profit	Carrying amount
			(In thousands of Korean won)			
Archigen Biotech Limited	₩ 155,901,033	50.00%	₩ 77,950,517	₩ -	₩ (29,118,224)	₩ 48,832,293
Samsung Bioepis	382,639,454	93.31%	357,049,036	4,538,474,961	-	4,895,523,997

- (*) Adjustment of fair value of net assets was applied at the point of loss of control.

Investee	December 31, 2015					
	Net assets	Percentage of ownership	Company's share of net assets	Investment difference	Elimination of unrealized profit	Carrying amount
			(In thousands of Korean won)			
Archigen Biotech Limited	₩ 116,158,497	50.00%	₩ 58,079,248	₩ -	₩ (29,118,224)	₩ 28,961,024
Samsung Bioepis	290,531,810	91.20%	264,965,011	4,543,613,356	-	4,808,578,367

12. PROPERTY, PLANT AND EQUIPMENT:

(1) Details of property, plant and equipment

Details of property, plant and equipment as of December 31, 2016 and 2015, are as follows:

		December 31, 2016		
		Acquisition cost	Accumulated depreciation	Book value
		(In thousands of Korean won)		
Buildings	₩	294,885,656	₩ (16,907,193)	₩ 277,978,463
Structures		7,479,077	(279,020)	7,200,057
Machinery		643,893,848	(82,443,594)	561,450,254
Vehicles		451,695	(346,133)	105,562
Other tangible assets		18,817,867	(7,297,897)	11,519,970
Construction in progress		232,705,460	-	232,705,460
	₩	1,198,233,603	₩ (107,273,837)	₩ 1,090,959,766

		December 31, 2015		
		Acquisition cost	Accumulated depreciation	Book value
		(In thousands of Korean won)		
Buildings	₩	280,346,385	₩ (9,612,440)	₩ 270,733,945
Structures		1,582,694	(92,043)	1,490,651
Machinery		245,802,276	(38,275,031)	207,527,245
Vehicles		336,828	(263,536)	73,292
Other tangible assets		22,095,877	(3,246,672)	18,849,205
Construction in progress		391,581,799	-	391,581,799
	₩	941,745,859	₩ (51,489,722)	₩ 890,256,137

(2) Changes in property, plant and equipment

For the years ended December 31, 2016 and 2015, the changes in property, plant and equipment are as follows:

		2016					
		Beginning balance	Acquisition	Disposals	Depreciation	Others	Ending balance
		(In thousands of Korean won)					
Buildings	₩	270,733,945	₩ 252,101	₩ -	₩ (7,294,753)	₩ 14,287,170	₩ 277,978,463
Structures		1,490,651	-	-	(186,977)	5,896,383	7,200,057
Machinery		207,527,245	183,595	-	(44,168,563)	397,907,977	561,450,254
Vehicles		73,292	11,666	-	(82,596)	103,200	105,562
Other tangible assets		18,849,205	1,888,977	(1,033)	(4,051,537)	(5,165,642)	11,519,970
Construction in progress		391,581,799	160,656,011	-	-	(319,532,350)	232,705,460
	₩	890,256,137	₩ 162,992,350	₩ (1,033)	₩ (55,784,426)	₩ 93,496,738	₩ 1,090,959,766

		2015						
		Beginning balance	Acquisition	Disposals	Depreciation	Others	Loss of control	Ending balance
		(In thousands of Korean won)						
Land	₩	28,063,117	₩ 4,655	₩ -	₩ -	₩ -	₩ (28,067,772)	₩ -
Buildings		88,550,264	1,629	-	(6,612,180)	209,247,847	(20,453,615)	270,733,945
Structures		1,853,044	-	-	(39,043)	482,808	(806,158)	1,490,651
Machinery		203,028,157	4,117,721	(601,858)	(23,270,692)	41,180,466	(16,926,549)	207,527,245
Vehicles		126,499	-	-	(84,207)	31,000	-	73,292
Other tangible assets		7,211,958	2,402,975	(1,888)	(2,688,515)	13,372,157	(1,447,482)	18,849,205
Construction in progress		432,461,197	226,242,465	-	-	(264,256,278)	(2,865,585)	391,581,799
	₩	761,294,236	₩ 232,769,445	₩ (603,746)	₩ (32,694,637)	₩ 58,000	₩ (70,567,161)	₩ 890,256,137

(3) Property, Plant and Equipment under construction

The Company recorded construction in progress in the amount of ₩162,992,350 thousand and ₩214,170,837 thousand to construct building and manufacturing facilities for the years ended December 31, 2016 and 2015. In addition, ₩5,921,756 thousand and ₩17,270,791 thousand of construction in progress has been added from borrowing costs capitalization for the years ended December 31, 2016 and 2015. Interest rates used to calculate the borrowing costs capitalization are 2.00%–4.16% and 1.67%–4.16% for the years ended December 31, 2016 and 2015, respectively.

13. INTANGIBLE ASSETS:

(1) Details of intangible assets

Details of intangible assets as of December 31, 2016 and 2015, are as follows:

December 31, 2016				
	Acquisition cost	Accumulated amortization	Government grants	Book value
(In thousands of Korean won)				
Software	₩ 19,864,004	₩ (9,254,120)	₩ -	₩ 10,609,884
Other intangible assets	3,054,155	(479,900)	-	2,574,255
Land use rights	69,525,088	(7,879,510)	(61,645,578)	-
Construction in progress	860,515	-	-	860,515
	<u>₩ 93,303,762</u>	<u>₩ (17,613,530)</u>	<u>₩ (61,645,578)</u>	<u>₩ 14,044,654</u>

December 31, 2015				
	Acquisition cost	Accumulated amortization	Government grants	Book value
(In thousands of Korean won)				
Software	₩ 13,493,581	₩ (5,044,457)	₩ -	₩ 8,449,124
Other intangible assets	2,906,501	(324,305)	-	2,582,196
Land use rights	69,525,088	(6,489,008)	(63,036,080)	-
Construction in progress	3,254,420	-	-	3,254,420
	<u>₩ 89,179,590</u>	<u>₩ (11,857,770)</u>	<u>₩ (63,036,080)</u>	<u>₩ 14,285,740</u>

(2) Changes in intangible assets

For the years ended December 31, 2016 and 2015, the changes in intangible assets are as follows:

	2016							
	Beginning balance	Acquisition	Disposals	Amortization	Amortization of government grants	Others	Ending balance	
	(In thousands of Korean won)							
Software	₩ 8,449,124	₩ 1,830,427	₩ -	₩ (4,209,869)	₩ -	₩ 4,540,202	₩ 10,609,884	
Other intangible assets	2,582,196	582,993	-	(155,594)	-	(435,340)	2,574,255	
Land use rights	-	-	-	(1,390,502)	1,390,502	-	-	
Construction in progress	3,254,420	2,765,706	-	-	-	(5,159,611)	860,515	
	₩ 14,285,740	₩ 5,179,126	₩ -	₩ (5,755,965)	₩ 1,390,502	₩ (1,054,749)	₩ 14,044,654	

2015									
	Beginning balance	Acquisition	Disposals	Amortization	Amortization of government grants	Others	Loss of control	Ending balance	
(In thousands of Korean won)									
Development									
Cost	₩ 292,046,534	₩ 141,913,740	₩ -	₩ (3,620,490)	₩ -	₩ -	₩(430,339,784)	₩ -	
Software	12,280,469	1,530,870	-	(3,595,176)	-	1,710,077	(3,477,116)	8,449,124	
Other intangible assets	7,404,236	4,050,868	(2,817,073)	(1,717,969)	1,390,502	479,400	(4,817,266)	2,582,196	
Land use rights	-	-	-	(1,390,502)	-	-	-	-	
Construction in progress	487,633	6,522,702	-	-	-	(2,247,477)	(1,508,438)	3,254,420	
Total	₩ 312,218,872	₩ 154,018,180	₩ (2,817,073)	₩ (10,324,137)	₩ 1,390,502	₩ (58,000)	₩ (440,142,604)	₩ 14,285,740	

- (3) The Company performed an impairment test on club memberships and development costs, which have indefinite useful lives. The recoverable amount of an asset or CGU is the higher of its value in use and its fair value less costs to sell. The value in use is estimated by applying a pretax discount rate that reflect current market assessments of the time value of money and the risks specific to the asset or CGU for which estimated future cash flows have not been adjusted, to the estimated future cash flows expected to be generated by the asset or CGU.

14. LOANS AND BORROWINGS:

- (1) Loans and borrowings as of December 31, 2016 and 2015, are as follows:

	December 31, 2016		December 31, 2015	
	Current	Non-current	Current	Non-current
(In thousands of Korean won)				
Bond issues	₩ 169,917,877	₩ 50,000,000	₩ -	₩ 50,000,000
Short-term borrowings	-	-	10,109,490	-
Long-term borrowings	233,776,207	389,363,194	-	631,792,000
	₩ 403,694,084	₩ 439,363,194	₩ 10,109,490	₩ 681,792,000

- (2) Details of bond issued as of December 31, 2016 and 2015, are as follows:

	The date of issue	The date of maturity	Interest rate (%)	December 31, 2016	December 31, 2015
(In thousands of Korean won)					
The 1 st placement bond	2014-10-24	2019-10-24	3.48	₩ 50,000,000	₩ 50,000,000
The 2 nd placement bond	2016-01-28	2017-01-28	2.38	39,995,381	-
The 3 rd placement bond	2016-03-31	2017-03-31	2.28	79,969,875	-
The 4 th placement bond	2016-08-12	2017-08-12	2.00	49,952,621	-
				₩ 219,917,877	₩ 50,000,000

- (3) Details of short-term borrowings as of December 31, 2016 and 2015, are as follows:

Financial institution	Description	Interest rate (%)	The date of maturity	December 31, 2016	December 31, 2015
(In thousands of Korean won)					
The Export-Import Bank of Korea	Loans for equipment	LIBOR + 1.90	2016-02-15	₩ -	₩ 5,512,609
		LIBOR + 1.90	2016-06-24	-	3,825,028
Mizuho Bank, Ltd	Loans for working	LIBOR + 1.04	2016-06-08	-	771,853
				₩ -	₩ 10,109,490

(4) Details of long-term borrowings as of December 31, 2016 and 2015, are as follows:

Financial institution	Description	Interest rate (%)	The date of maturity	December 31, 2016	December 31, 2015
(In thousands of Korean won)					
The Korea Development Bank	Loans for equipment	3.79–4.00	2017-08-21	₩ 50,000,000	₩ 50,000,000
		3.24–3.57	2019-07-24	50,000,000	50,000,000
		2.86–2.90	2020-06-12 (*1)	50,000,000	50,000,000
		2.86	2020-12-15 (*1)	40,000,000	40,000,000
The Export-Import Bank of Korea	Loans for import	LIBOR + 1.90	2017-06-23	7,251,000	7,032,000
	Loans for investment	LIBOR + 1.50	2019-06-14	16,314,750	-
Kookmin Bank	General loan	4.16	2016-12-02	-	41,000,000
		CD (91 days)			
		+1.54	2020-08-11 (*2)	50,000,000	50,000,000
Hana Bank		CD (91 days)			
		+1.45	2019-11-14	100,000,000	100,000,000
		CD (91 days)			
		+1.60	2021-02-26 (*3)	800,000	-
Nonghyup Bank		MOR (1 year)			
		+1.52	2019-11-25	50,000,000	50,000,000
Shinhan Bank (*)	Loans for equipment	3.98	2017-01-02	100,000,000	100,000,000
Mizuho Bank, Ltd	Foreign currency loans	LIBOR+1.35	2018-02-12	8,879,885	8,611,689
		LIBOR+1.35	2018-05-23	5,359,472	5,197,601
		LIBOR+1.35	2018-06-26	1,672,326	1,621,817
		LIBOR+1.35	2018-07-23	1,557,501	1,510,460
		LIBOR+1.35	2018-08-03	2,685,608	2,604,495
		LIBOR+1.35	2017-11-28	31,898,928	30,935,493
		LIBOR+1.35	2017-12-26	8,371,279	8,118,445
Australia New Zealand Bank		LIBOR+1.85	2017-04-28	11,686,195	11,333,240
		LIBOR+1.85	2017-07-25	12,459,635	12,083,320
		LIBOR+1.85	2017-09-22	12,109,170	11,743,440
Industrial & Commercial Bank of China Ltd		LIBOR+1.50	2019-06-29	12,093,652	-
Transfer of current portion				(233,776,207)	-
				₩ 389,363,194	₩ 631,792,000

(*1) The loan has the condition for level payment every three months, two years before its maturity.

(*2) The loan has the condition for level payment every three months from November 11, 2018, to August 11, 2020.

(*3) The loan has the condition for level payment every three months from May 27, 2019, to February 26, 2021.

(5) The repayment schedule of bond and borrowings as of December 31, 2016, is as follows:

Period	Bond issues	Long-term borrowings	Total
(In thousands of Korean won)			
2017.01.01–2017.12.31	₩ 170,000,000	₩ 233,776,207	₩ 403,776,207
2018.01.01–2018.12.31	-	38,904,793	38,904,793
2019.01.01–2019.12.31	50,000,000	298,708,402	348,708,402
2020.01.01–2020.12.31	-	51,650,000	51,650,000
2021.12.31–	-	100,000	100,000
	₩ 220,000,000	₩ 623,139,402	₩ 843,139,402

15. TRADE AND OTHER PAYABLES:

Details of trade and other payables as of December 31, 2016 and 2015, are as follows:

	December 31, 2016		December 31, 2015	
	Current	Non-current	Current	Non-current
	(In thousands of Korean won)			
Trade payables	₩ 1,521,209	₩ -	₩ 358,223	₩ -
Non-trade payables	113,489,469	-	27,495,078	-
Accrued expenses	23,045,697	2,300,380	28,340,590	1,909,574
	<u>₩ 138,056,375</u>	<u>₩ 2,300,380</u>	<u>₩ 56,193,891</u>	<u>₩ 1,909,574</u>

16. OTHER LIABILITIES:

Details of other current liabilities as of December 31, 2016 and 2015, are as follows:

	December 31, 2016		December 31, 2015	
	Current	Non-Current	Current	Non-Current
	(In thousands of Korean won)			
Advance receipts	₩ 58,708,693	₩ -	₩ 23,965,072	₩ -
Withholdings	1,615,098	-	961,701	-
	<u>₩ 60,323,791</u>	<u>₩ -</u>	<u>₩ 24,926,773</u>	<u>₩ -</u>

17. EMPLOYEE BENEFITS:

(1) Recognized liabilities for employee benefits as of December 31, 2016 and 2015, are as follows:

		December 31, 2016	December 31, 2015
Explanation		(In thousands of Korean won)	
Defined benefit obligations	Present value of defined benefit obligations	₩ 23,333,875	₩ 15,502,178
	Fair value of plan assets	(17,576,610)	(6,787,823)
		<u>5,757,265</u>	<u>8,714,355</u>
Trade and other payables	Accrued expenses(*)	15,670,445	9,163,282
	Long-term accrued expenses(*)	2,300,380	1,909,574
		<u>17,970,825</u>	<u>11,072,856</u>
		<u>₩ 23,728,090</u>	<u>₩ 19,787,211</u>

(*) The amounts include the annual leave, performance bonuses and others paid to employees.

(2) Changes in the present value of defined benefit obligation and plan assets for the years ended December 31, 2016 and 2015, are as follows:

	2016		
	Defined benefit obligations	Plan assets	Net defined benefit liability
	(In thousands of Korean won)		
Beginning balance at January 1	₩ 15,502,179	₩ (6,787,823)	₩ 8,714,356
Included in profit or loss:			
Current service cost	6,112,437	-	6,112,437
Interest expenses (income)	470,654	(206,082)	264,572
	<u>6,583,091</u>	<u>(206,082)</u>	<u>6,377,009</u>
Included in other comprehensive income:			
Remeasurement losses (gains):	1,952,438	-	1,952,438
Actuarial losses (gains) arising from demographic assumptions	(181,982)	-	(181,982)
Financial assumptions	(135,755)	-	(135,755)
Experience adjustment	2,270,175	-	2,270,175
Return on plan assets excluding interest income	-	153,008	153,008
	<u>1,952,438</u>	<u>153,008</u>	<u>2,105,446</u>
Other :			
Contributions paid into the plan	-	(11,056,300)	(11,056,300)
Benefits paid by the plan	(703,833)	320,587	(383,246)
Ending balance at December 31	<u>₩ 23,333,875</u>	<u>₩ (17,576,610)</u>	<u>₩ 5,757,265</u>

	2015		
	Defined benefit obligations	Plan assets	Net defined benefit liability
	(In thousands of Korean won)		
Beginning balance at January 1	₩ 15,769,625	₩ (7,311,864)	₩ 8,457,761
Included in profit or loss:			
Current service cost	5,587,815	-	5,587,815
Interest expenses (income)	510,851	(204,020)	306,831
	<u>6,098,666</u>	<u>(204,020)</u>	<u>5,894,646</u>
Included in other comprehensive income:			
Remeasurement losses (gains):	4,446,043	-	4,446,043
Actuarial losses (gains) arising from demographic assumptions	291,955	-	291,955
Financial assumptions	1,284,573	-	1,284,573
Experience adjustment	2,869,515	-	2,869,515
Return on plan assets, excluding interest income	-	65,384	65,384
	<u>4,446,043</u>	<u>65,384</u>	<u>4,511,427</u>
Other :			
Contributions paid into the plan	-	(5,517,800)	(5,517,800)
Benefits paid by the plan	(1,067,689)	999,588	(68,101)
Transfer in from related parties	1,092,019	(410,573)	681,446
Loss of control	(10,836,486)	5,591,462	(5,245,024)
Ending balance at December 31	<u>₩ 15,502,178</u>	<u>₩ (6,787,823)</u>	<u>₩ 8,714,355</u>

(3) The plan assets as of December 31, 2016 and 2015, are as follow:

		<u>December 31, 2016</u>		<u>December 31, 2015</u>
		(In thousands of Korean won)		
Cash and cash equivalents	₩	17,576,610	₩	6,787,823

(4) Principal actuarial assumption as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
Discount rate for defined benefit obligations	3.30%	3.30%
Future salary increases	5.91%	5.74%

In order to calculate present value of defined benefit obligations, the Company determined discount rate based on return of corporate bonds, which are consistent with currency of defined benefit obligations and expected payment term as of the reporting date.

(5) As of December 31, 2016, effects on present value of defined benefit obligations if each of significant actuarial assumptions changes within potential reasonable range are as follows (if changed by 1%):

		<u>1% Increase</u>		<u>1% Decrease</u>
		(In thousands of Korean won)		
Discount rate	₩	(1,673,359)	₩	1,970,794
Expected rate of salary increase	₩	1,969,478	₩	(1,702,828)

(6) As of December 31, 2016, the weighted-average duration of the defined benefit obligation is 7.53 years.

18. CAPITAL:

(1) Details of share capital as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In Korean won and shares)	
Number of shares authorized	500,000,000	100,000,000
Value per share	2,500	5,000
Number of shares issued	66,165,000	27,568,721
Share capital	₩165,412,500,000	₩ 137,843,605,000

(2) Change in number of shares issued for the years ended December 31, 2016 and 2015, are as follows:

	<u>2016</u>		<u>2015</u>	
	<u>Number of Shares issued</u>	<u>Outstanding Stock</u>	<u>Number of Shares issued</u>	<u>Outstanding Stock</u>
Beginning number of shares	27,568,721	27,568,721	20,366,649	20,366,649
Split value	27,568,721	27,568,721	-	-
Capital increase	11,027,558	11,027,558	7,202,072	7,202,072
Ending number of shares	<u>66,165,000</u>	<u>66,165,000</u>	<u>27,568,721</u>	<u>27,568,721</u>

(3) Details of share premium as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In thousands of Korean won)	
Additional paid-up capital	₩ 2,496,583,409	₩ 1,039,890,932
Changes in equity due to(*) irregular capital increase	(9,270,327)	(9,270,327)
	<u>₩ 2,487,313,082</u>	<u>₩ 1,030,620,605</u>

(*) Due to the capital increase, the amount of additional paid-up capital increased by ₩1,456,692,477 thousand.

(4) Details of accumulated other comprehensive income (loss) as of December 31, 2016 and 2015, are as follows:

	<u>December 31, 2016</u>	<u>December 31, 2015</u>
	(In thousands of Korean won)	
Defined benefit plan actuarial losses	₩ (6,420,506)	₩ (4,824,578)
Net change in other comprehensive income of equity-accounted investees	11,367,511	9,664,244
	<u>₩ 4,947,005</u>	<u>₩ 4,839,666</u>

(5) Appropriations of retained earnings as of December 31, 2016 and 2015, is as follows:

STATEMENTS OF APPROPRIATIONS OF RETAINED EARNINGS
FOR THE YEARS ENDED DECEMBER 31, 2016 AND 2015

	<u>2016</u>	<u>2015</u>
	(In Korean won)	
RETAINED EARNINGS BEFORE APPROPRIATIONS:		
Unappropriated retained earnings (undisposed deficit) carried over from prior	₩ 1,601,530,282,540	₩ (133,258,883,219)
Effect of changing consolidation scope	-	(170,156,750,840)
Net income (loss)	(176,823,409,527)	1,904,945,916,599
	<u>1,424,706,873,013</u>	<u>1,601,530,282,540</u>
APPROPRIATIONS OF RETAINED EARNINGS	-	-
UNAPPROPRIATED RETAINED EARNINGS TO BE CARRIED FORWARD TO SUBSEQUENT YEAR	<u>₩ 1,424,706,873,013</u>	<u>₩ 1,601,530,282,540</u>

19. REVENUE:

Details of revenue for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Product revenue	₩ 240,820,443	₩ 40,013,975
Other revenue	53,801,578	51,264,054
	<u>₩ 294,622,021</u>	<u>₩ 91,278,029</u>

20. COST OF SALES:

Details of cost of sales for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Cost of product sales	₩ 199,420,965	₩ 46,205,836
Cost of other sales	68,667,393	68,578,781
	<u>₩ 268,088,358</u>	<u>₩ 114,784,617</u>

21. SELLING, GENERAL AND ADMINISTRATIVE EXPENSES:

Details of selling, general and administrative expenses for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Salaries	₩ 11,120,503	₩ 33,143,488
Bonuses	5,005,847	13,622,585
Retirement benefits	1,087,455	3,049,972
Welfare expenses	4,322,572	9,982,951
Depreciation	4,246,229	11,576,657
Amortization	1,829,666	4,036,852
Repair expenses	1,008,278	2,030,329
Supplies expenses	832,904	2,225,601
Utility expenses	207,601	612,501
Travel expenses	1,441,789	6,230,363
Service fees	11,901,152	66,672,830
Training expenses	5,498,193	6,166,715
Electronic processing expenses	878,147	5,285,461
Rental expenses	235,209	2,960,454
Taxes and dues	1,200,086	4,555,343
Others	6,140,671	7,983,754
	<u>₩ 56,956,302</u>	<u>₩ 180,135,856</u>

22. NATURE OF EXPENSES:

Details of nature of expenses for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Changes in inventories	₩ (21,633,321)	₩ (44,774,933)
Used raw materials	24,934,106	21,268,810
Salaries and bonuses	110,259,584	87,269,318
Depreciation and amortization	60,149,889	41,628,273
Supplies expenses	57,984,312	21,795,354
Utility expenses	14,699,093	612,501
Service fees	27,928,209	80,486,353
Others	50,722,788	86,634,797
	₩ 325,044,660	₩ 294,920,473

23. OTHER INCOME AND EXPENSES:

Details of other income and other expenses for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Other income		
Rental revenue	₩ 562,290	₩ 62,652
Fees revenue	205,488	51,716
Gain on disposal of intangible assets	-	682,927
Miscellaneous revenue	450,052	1,866,286
	1,217,830	2,663,581
Other expenses		
Donations	24,343	202,384
Loss on disposal of property, plant and equipment	1,033	583,986
Miscellaneous losses	13,634	34,883
	₩ 39,010	₩ 821,253

24. FINANCIAL INCOME AND EXPENSES:

Details of financial income and expenses for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Financial income		
Interest income	₩ 3,639,650	₩ 1,182,311
Gain on foreign currency translations	2,825,232	554,928
Gain on foreign currency transactions	7,883,685	5,107,485
	14,348,567	6,844,724
Financial expenses		
Interest expenses	19,254,372	2,650,470
Loss on foreign currency translations	3,962,749	9,188,935
Loss on foreign currency transactions	9,534,066	8,918,343
Loss on valuation of derivatives	54,031,894	1,820,439,050
	₩ 86,783,081	₩ 1,841,196,798

25. INCOME TAX EXPENSE:

(1) Income tax expense for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Current tax expense:		
Current tax expense	₩ -	₩ 1,969
Deferred tax expense		
Temporary differences	(54,903,810)	581,574,424
Income tax recognized in other comprehensive income	826,237	1,540,300
Total	₩ (54,077,573)	₩ 583,114,724
Income tax expense (profit)	₩ (54,077,573)	₩ 583,116,693

(2) The relationship between tax expense and accounting profit for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Loss before income taxes	₩ (230,900,983)	₩ 2,488,062,610
Tax using the Company's domestic tax rate	(55,878,038)	601,649,152
Adjustment:		
Non-deductible expense	851,359	103,471
Recognition of tax effect of previously unrecognized tax	-	(52,160,122)
Increase in unrecognized deferred tax assets	949,106	32,535,749
Others	-	988,443
Income tax expense	₩ (54,077,573)	₩ 583,116,693
Effective income tax rate (*)	-	23.44%

(*) Effective tax rate was not calculated since the Company recorded loss before income tax expense for the year ended December 31, 2016.

(3) Income taxes recognized directly in equity for the years ended December 31, 2016 and 2015, are as follows:

	2016		2015	
	Before tax	Tax expense	Net of tax	
	(In thousands of Korean won)			
Remeasurement of defined benefit plans	₩ (2,105,447)	₩ 509,519	₩ (1,595,928)	
Change on equity method	1,386,549	316,718	1,703,267	
	₩ (718,898)	₩ 826,237	₩ 107,339	
	2015		2015	
	Before tax	Tax expense	Net of tax	
	(In thousands of Korean won)			
Remeasurement of defined benefit plans	₩ (4,511,427)	₩ 1,540,300	₩ (2,971,127)	

- (4) Changes in temporary differences and deferred tax assets (liabilities) as of December 31, 2016 and 2015, are as follows:

	2016		
	Beginning balance	Change in	Ending balance
	(In thousands of Korean won)		
Accrued expenses	₩ 8,938,128	₩ (124,623)	₩ 8,813,505
Net defined benefit obligations	7,345,550	(2,573,843)	4,771,707
Long-term accrued expenses	1,909,574	390,805	2,300,379
Loss (gain) on foreign currency translations	5,462,583	5,335,387	10,797,970
Provision for loss on valuation of finished goods	11,271,591	(2,858,375)	8,413,216
Depreciation	-	216,563	216,563
Provision for performance bonus	6,120,438	5,320,114	11,440,552
Derivative liabilities	1,820,439,050	54,031,894	1,874,470,944
Equity-accounted investees	(4,187,771,339)	127,836,101	(4,059,935,238)
Provision for loss on valuation of work in progress	842,760	1,582,983	2,425,743
Accrued income	-	(2,668,507)	(2,668,507)
Tax loss carryforward	<u>145,269,583</u>	<u>44,308,670</u>	<u>189,578,253</u>
	<u>(2,180,172,082)</u>	<u>230,797,169</u>	<u>(1,949,374,913)</u>
Tax credit carryforward	32,140,305	2,628,074	34,768,379
Deferred tax liabilities (assets)	<u>₩ 581,574,424</u>		<u>₩ 526,670,614</u>

	2015		
	Beginning balance	Change in	Ending balance
	(In thousands of Korean won)		
Accrued expenses	₩ -	₩ 8,938,128	₩ 8,938,128
Net defined benefit obligations	-	7,345,550	7,345,550
Long-term accrued expenses	-	1,909,574	1,909,574
Loss (gain) on foreign currency translations	-	5,462,583	5,462,583
Provision for loss on valuation of finished goods	-	11,271,591	11,271,591
Provision for performance bonus	-	6,120,438	6,120,438
Derivative liabilities	-	1,820,439,050	1,820,439,050
Equity-accounted investees	-	(4,187,771,339)	(4,187,771,339)
Provision for loss on valuation of work in progress	-	842,760	842,760
Tax loss carryforward	<u>-</u>	<u>145,269,583</u>	<u>145,269,583</u>
	<u>-</u>	<u>(2,180,172,082)</u>	<u>(2,180,172,082)</u>
Tax credit carryforward	-	32,140,305	32,140,305
Deferred tax liabilities (assets)	<u>₩ -</u>		<u>₩ 581,574,424</u>

Deferred tax assets that not recognized because of realization with respect to equity-accounted investees and Tax credit carryforward as of December 31, 2016 and 2015, are ₩89,690,264 thousand and ₩86,113,085 thousand, respectively.

26. EARNINGS (LOSS) PER SHARE:

(1) Basic earnings (loss) per share for the years ended December 31, 2016 and 2015, are calculated as follows:

	2016	2015
	(In Korean won)	
Profit (loss) for the year, attributable to the owners of the Company	₩ (176,823,409,527)	₩ 1,920,178,773,938
Weighted-average number of ordinary shares	<u>56,764,459</u>	<u>49,453,672</u>
Basic earnings (loss) per share	₩ (3,115)	₩ 38,828

(2) Calculation for weighted-average number of common shares outstanding for the years ended December 31, 2016 and 2015, are as follow:

Description	2016			
	Share	Period	Days	Multiplied
Beginning of number of issued common stock	27,568,721	2016.01.01–2016.12.31	366	10,090,151,886
Split value	27,568,721	2016.01.01–2016.12.31	366	10,090,151,886
Capital increase	<u>11,027,558</u>	2016.11.08–2016.12.31	54	595,488,132
	<u>66,165,000</u>			<u>20,775,791,904</u>
Days				366
Weighted-average number of ordinary shares				56,764,459

Description	2015			
	Share(*)	Period	Days	Multiplied
Beginning of number of issued common stock	40,733,298	2015.01.01–2015.12.31	365	14,867,653,770
Capital increase	7,077,860	2015.02.24–2015.12.31	311	2,201,214,460
Capital increase	<u>7,326,284</u>	2015.08.20–2015.12.31	134	<u>981,722,056</u>
	<u>55,137,442</u>			<u>18,050,590,286</u>
Days				365
Weighted-average number of ordinary shares				49,453,672

(*) The number of shares from previous year has been retroactively adjusted due to split value.

(3) Diluted earnings (loss) per share

The Company has no dilutive potential shares; therefore, diluted earnings (loss) per share is equal to the basic earnings (loss) per share.

27. COMMITMENTS AND CONTINGENCIES:

(1) Biogen Inc. has an option to purchase additional shares of Samsung Bioepis Co., Ltd. that would allow them to increase their ownership percentage up to 49.9%. The option will expire in 2018, and the option price is the sum of principal amount and accumulated interest. The value of the financial derivative liabilities may be changed significantly depending on the market situation and the exercise of the option. If Biogen Inc. does not exercise the option until the option expires in 2018, the determination of control could change and it could result in a loss (see Notes 5 and 11).

(2) Details of the Company's commitments for loans with financial institutions as of December 31, 2016, are as follows:

Financial institution	Remarks	Limit	Outstanding balance
(In thousands of won, USD)			
Woori Bank	Bank overdraft	₩ 10,000,000	₩ -
	B2B PLUS	20,000,000	-
KEB Hana Bank	Bank overdraft	10,000,000	-
	Loans for equipment	100,000,000	100,000,000
	Loans for equipment	50,000,000	800,000
	Comprehensive limit	40,000,000	-
Shinhan Bank	General loan	19,214,000	-
	Bank overdraft	10,000,000	-
	Loans for equipment	100,000,000	100,000,000
Kookmin Bank	Loans for equipment	50,000,000	50,000,000
NH Nonghyup Bank	Loans for equipment	50,000,000	50,000,000
The Korea Development Bank	Loans for equipment	190,000,000	190,000,000
The Export-Import Bank of Korea	Loans for equipment in foreign currency	USD 6,000	USD 6,000
	Loans for investment in foreign currency	USD 13,500	USD 13,500
Australia and New Zealand Bank	Loans for equipment in foreign currency	USD 30,000	USD 30,000
Mizuho Bank, Ltd	Loans for equipment in foreign currency	USD 50,000	USD 50,000
	General loan in foreign currency	USD 10,000	-
Industrial and Commercial Bank of China Limited	Loans for equipment in foreign currency	USD 50,000	USD 10,007

(3) Details of guarantees provided by financial institutions as of December 31, 2016, are as follows:

Financial institution	Remarks	Limit	Outstanding balance
(In thousands of won, USD)			
Woori Bank	Payment guarantees	KRW 9,500,000	KRW 5,501,579
	Letter of credit	USD 8,500	USD -
	Mortgage-backed payment guarantee	USD 17,800	USD -
Shinhan Bank	Letter of credit	USD 10,000	USD -
	Foreign currency payment guarantee	USD 10,000	USD -
Kookmin Bank	Foreign currency payment guarantee	KRW 30,000,000	USD 25,000

(4) Major commitments with financial institutions

If the Company did not comply with the commitments below, they may be required the early repayment in long-term borrowings. Details of covenants related to long-term borrowings with financial institutions as of December 31, 2016, are as follows:

Financial institution	Remarks	Details of covenants
Mizuho Bank, Ltd	Financial Covenant	The Company shall maintain a maximum ratio of Interest-Bearing Debt to Shareholders' Equity of 100%.
	Shareholding of Borrower	The Company shall continue to belong to the Samsung Group (as defined under the Monopoly Regulation and Fair Trade Act of Korea). Samsung Electronics Co., Ltd. and Samsung C&T Corporation shall collectively maintain over 50% shareholding in the Company.
Australia and New Zealand Bank	Financial Covenant	The Company shall maintain a maximum ratio of Interest-Bearing Debt to Shareholders' Equity of 200%.

- (5) In the process of concluding a lease agreement with Incheon Metropolitan city (“Incheon”) related to the use of land, the Company is exempt from the full amount of rent because the Company meets the requirements for exemption of rent in accordance with the Common Property Management Ordinance of Incheon. The Company recognized as intangible assets (government grant) at the fair value of the rent that are exempt from Incheon, which the Company recognized as a reduction of assets. The period of land use without payment and exemption requirement of the full amount of rent are as follows:

1) Period

Lease terms are for a total of 50 years. The initial lease period is for 20 years from the land lease contract date, May 1, 2011, and the Company can renew the contract for another 30 years in successive 10-year increments.

2) Exemption requirement

The Company whose foreign investment amounts to more than USD 20 million is subject to the exemption from Common Property Management Ordinance of Incheon. After that, the Company, including foreign invested enterprises that meet the average daily employee count of above 300, is subject to exemption.

28. RELATED PARTIES:

- (1) As Samsung C&T Corporation and Cheil Industries Inc., merged in the prior period, Samsung C&T Corporation (formerly, Cheil Industries Inc.) became the Company’s controlling company. As of December 31, 2016, the percentage of ownership Samsung C&T Corporation (the former Cheil Industries Inc.) has over the Company is 43.44%.
- (2) As of December 31, 2016, the Company's related parties consist of ① Samsung C&T Corporation and Subsidiaries (Controlling company), ② Samsung Electronics Co., Ltd. (Entity which has significant influence) ③ Samsung Bioepis Co., Ltd. and Subsidiaries (Equity-accounted investee) and Archigen Biotech Limited (Joint venture). Meanwhile, the Enterprise Group, which the Group belongs to, in accordance with the Monopoly Regulation and Fair Trade Act of the Republic of Korea, consists of 58 entities including Samsung Electronics Co., Ltd., as of December 31, 2016.
- (3) Details of the associates and other related parties who have transactions or account balances with the Company as of December 31, 2016, are as follows:

Description	Related party
Parent	Samsung C&T Corporation
Entity which has significant influence	Samsung Electronics Co., Ltd.
Associates and joint ventures	Samsung Bioepis Co., Ltd. Archigen Biotech Limited
Subsidiaries of parent	Samsung Welstory Incorporated, Samsung C&T Deutschland GmbH
The Enterprise Group (*)	Samsung Card Co., Ltd. Samsung Engineering Co., Ltd. S-I Corporation Samsung SDS Co., Ltd. Samsung Economic Research Institute Multicampus Corporation, Hotel Shilla Co., Ltd. Samsung Fire & Marine Insurance Co., Ltd. Samsung Securities Co., Ltd. Samsung Life Insurance Co., Ltd. Cheil Worldwide Inc., Samsung Fine Chemicals Co., Ltd.

(4) Significant receivables and payables with related parties as of December 31, 2016 and 2015, are as follows:

		December 31, 2016					
		Receivables				Payables	
		Other		Accrued		Other	
Description	Related party	receivables		income		payables	
(In thousands of Korean won)							
Parent	Samsung C&T Corporation	₩	-	₩	-	₩	13,484
Entity which has significant influence	Samsung Electronics Co., Ltd.		-		-		18,270
Associates and joint ventures	Samsung Bioepis Co., Ltd.		78,972		-		-
	Archigen Biotech Limited		10,632		19,678		-
Subsidiaries of parent	Samsung Welstory Incorporated		10,604		-		512,524
	Samsung C&T Deutschland GmbH		-		-		74,659
		₩	100,208	₩	19,678	₩	618,937
		December 31, 2015					
		Receivables				Payables	
		Other		Accrued		Other	
Description	Related party	receivables		income		payables	
(In thousands of Korean won)							
Parent	Samsung C&T Corporation	₩	-	₩	-	₩	3,373,788
Entity which has significant influence	Samsung Electronics Co., Ltd.		-		-		941
Associates and joint ventures	Samsung Bioepis Co., Ltd.(*)		83,080		-		-
	Archigen Biotech Limited		-		4,809		-
Subsidiaries of parent	Samsung Welstory Incorporated		-		-		351,510
		₩	83,080	₩	4,809	₩	3,726,239

(*) Gain or loss on transactions before the date of losing control are included in the statement of comprehensive income.

(5) Significant receivables and payables with the Enterprise Group except for the transactions included in "Transaction with related parties" as of December 31, 2016 and 2015, are as follows:

Description	Related party	December 31,	December 31,
		2016	2015
(In thousands of Korean won)			
The Enterprise Group	Samsung Card Co., Ltd.	₩ 67,292,579	₩ 5,859,369
	Samsung Engineering Co., Ltd.	26,901,600	-
	S-1 Corporation	1,058,094	614,388
	Samsung SDS Co., Ltd.	674,323	1,804,338
	Samsung Economic Research Institute	98,182	44,293
	Multicampus Corporation	8,364	30,860
	Hotel Shilla Co., Ltd.	4,982	-
	Samsung Fire & Marine Insurance Co., Ltd.	432	323
		₩ 96,038,556	₩ 8,353,571

- (6) Significant transactions that occurred in the normal course of business with related parties for the years ended December 31, 2016 and 2015, are as follows:

		2016							
		Revenue				Expense			
Description	Related party	Commission	Others		Service fee		Others		
			(In thousands of Korean won)						
Parent	Samsung C&T Corporation	₩	-	₩	-	₩	143,846	₩	23,035
Entity which has significant influence	Samsung Electronics Co., Ltd.								
			-		-		-		18,059
Associates and joint ventures	Samsung Bioepis Co., Ltd.(*1)		205,488		452,406		-		-
	Archigen Biotech Limited		-		329,879		-		-
Subsidiaries of parent	Samsung Welstory Incorporated.		-		37,686		4,196		4,978,265
	Samsung C&T Deutschland GmbH		-		-		367,481		15,503
		₩	205,488	₩	819,971	₩	515,523	₩	5,034,862

		2015											
		Revenue				Expense							
Description	Related party	Other revenue		Others		Research and development (*2)		Service fee		Others		Asset acquisition	
(In thousands of Korean won)													
Parent	Samsung C&T Corporation	₩	-	₩	-	₩	-	₩	1,327,318	₩	202,592	₩	26,537,000
Entity which has significant influence	Samsung Electronics Co., Ltd.		-		-		-		6,241,452		12,725		-
Joint ventures	Archigen Biotech Limited		-		249,865		-		-		-		-
Subsidiaries of parent	Samsung Welstory Incorporated.		-		19,917		-		1,540,756		3,503,459		-
	Biogen Inc.(*1)		632,895		1,050,660		106,419,376		19,557,143		-		-
	Samsung C&T Deutschland GmbH		-		-		-		-		34,730		-
		₩	632,895	₩	1,320,442	₩	106,419,376	₩	28,666,669	₩	3,753,506	₩	26,537,000

(*1) Gain or loss on transactions with Biogen Inc. before the date of losing control over Samsung Bioepis Co., Ltd. Because the Company lost the control over Samsung Bioepis Co., during 2015, Biogen Inc. is no longer a related party.

(*2) Expenditures for acquiring inventories and intangible assets

- (7) Significant transactions that occurred in the normal course of business with the Enterprise Group except for the transactions included in "Transaction with related parties" for the years ended December 31, 2016 and 2015, are as follows:

		2016			
		Revenue		Expense	
Description	Related party	Others	Service fee	Others	Asset acquisition
(In thousands of Korean won)					
the Enterprise Group					
	Samsung Card Co., Ltd.	₩ -	₩ -	₩ 256,607	₩ -
	Samsung Engineering Co., Ltd.	-	-	-	161,183,000
	S-1 Corporation	-	5,242,071	392,710	556,150
	Samsung SDS Co., Ltd.	-	48,000	7,742,700	1,778,750
	Samsung Economic Research Institute	-	197,271	1,293,383	-
	Multicampus Corporation	-	8,000	365,184	-
	Hotel Shilla Co., Ltd.	-	47,416	54,674	14,269
	Samsung Fire & Marine Insurance Co., Ltd.	130,664	2,089	3,494,792	-
	Samsung Securities Co., Ltd.	-	-	2,699,539	-
	Samsung Life Insurance Co., Ltd.	2,640	-	-	-
	Cheil Worldwide Inc.	-	-	105,724	-
	Total	₩ 133,304	₩ 5,544,847	₩ 16,405,313	₩ 163,532,169

		2015			
		Revenue		Expense	
Description	Related party	Others	Service fee	Others	Asset acquisition
(In thousands of Korean won)					
the Enterprise Group					
	Samsung Card Co., Ltd.	₩ -	₩ -	₩ 410,241	₩ -
	Samsung Engineering Co., Ltd.	-	-	-	54,994,700
	S-1 Corporation	-	4,935,745	331,994	608,990
	Samsung SDS Co., Ltd.	-	773,604	9,602,467	4,943,560
	Samsung Economic Research Institute	-	30,310	1,381,718	-
	Multicampus Corporation	-	6,000	300,610	-
	Hotel Shilla Co., Ltd.	-	10,919	123,198	47,044
	Samsung Fire & Marine Insurance Co., Ltd.	103,320	3,055	4,258,069	-
	Samsung Life Insurance Co., Ltd.	86,238	16,405	-	-
	Cheil Worldwide Inc.	-	-	17,589	-
	Samsung Fine Chemicals Co., Ltd.	-	-	62,675	-
	Total	₩ 189,558	₩ 5,776,038	₩ 16,488,561	₩ 60,594,294

- (8) Key management personnel compensation in total and for each of the following categories for the years ended December 31, 2016 and 2015, are as follows:

	2016	2015
	(In thousands of Korean won)	
Short-term employee benefits	₩ 2,200,739	₩ 4,395,011
Long-term employee benefits	1,094,035	1,463,393
Retirement benefits	463,960	780,964
	₩ 3,758,734	₩ 6,639,368

Key management refers to the registered directors and auditor. Compensation of the Company's key management personnel includes salaries, non-cash benefits and contributions to a retirement benefit plan.

- (9) The Company made an additional investment to the associates and joint ventures, Samsung Bioepis Co., Ltd., and Archigen Biotech Limited, increased by ₩200,000 million and ₩34,653 million in 2016.
- (10) The Company concluded a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land, and subleased the land to Samsung Bioepis Co., Ltd., the associate, until the expiration date with the agreement of Incheon.

29. STATEMENTS OF CASH FLOWS:

Significant transactions not involving cash flows of investing activities for the years ended December 31, 2016 and 2015, are as follows:

		2016	2015
		(In thousands of Korean won)	
Changes in other payables due to property, plant and equipment	₩	86,508,215 ₩	(7,369,447)

30. FINANCIAL RISK MANAGEMENT:

The Company is exposed to credit risk, liquidity risk and market risk in connection with its financial instruments. This note presents information about the Company's exposure to each of the above risks, the Company's objectives, policies and processes for measuring and managing risks, and the Company's management of capital. Further quantitative disclosures are included throughout these financial statements.

(1) Financial risk management

The Company's risk management policies are established to identify and analyze the risks faced by the Company, to set appropriate risk limits and controls and to monitor risks and adherence to limits. Risk management policies and systems are reviewed regularly to reflect changes in market conditions and the Company's activities. The Company, through its training and management standards and procedures, aims to maintain a disciplined and constructive control environment in which all employees understand their roles and obligations. The Board of Directors oversees how management monitors compliance with the Company's risk management policies and procedures, and reviews the adequacy of the risk management framework in relation to the risks faced by the Company.

(2) Credit risk

Credit risk is the risk of financial loss to the Company if a customer or counterparty to a financial instrument fails to meet its contractual obligations, and arises principally from the Company's receivables from customers.

The carrying amount of financial assets represents the maximum credit exposure. The maximum exposure to credit risk as of December 31, 2016 and 2015, are as follows:

		December 31, 2016		December 31, 2015
		(In thousands of Korean won)		
Cash and cash equivalents	₩	230,337,601	₩	34,510,062
Short-term financial instruments		970,000,000		-
Long-term financial instruments		8,000		8,000
Trade and other receivables		75,630,820		37,009,243
Deposits provided		37,764,300		24,980,000
	₩	<u>1,313,740,721</u>	₩	<u>96,507,305</u>

(3) Liquidity risk

Liquidity risk is the risk that the Company will encounter difficulty in meeting the obligations associated with its financial liabilities that are settled by delivering cash or another financial asset. The Company's approach to managing liquidity is to ensure, as far as possible, that it will always have sufficient liquidity to meet its liabilities when they are due, under both normal and stressed conditions, without incurring unacceptable losses or risking damage to the Company's reputation.

The Company monitors its cash flows through long-term and short-term management strategies and ensures it has sufficient cash on demand to meet expected operational expenses. This excludes the potential impact of extreme circumstances that cannot reasonably be predicted.

The Company establishes short-term and long-term cash management plans to manage liquidity risk. The Company matches maturity structures of financial assets and liabilities through analyzing and reviewing cash flow budget and actual cash flow. Management believes that the Company is able to redeem its financial liabilities through operating cash flows and cash inflows of financial assets.

The following are the contractual maturities of financial liabilities as of December 31, 2016 and 2015, including estimated interest payments and excluding the impact of netting agreements. The Company does not expect that the cash flows included in the maturity analysis could occur significantly earlier or at different amounts.

December 31, 2016					
	Carrying amount	Contractual cash flows	Less than six months	6-12 months	More than one year
(In thousands of Korean won)					
Trade and other payables	₩ 140,356,755	₩ 140,356,755	₩ 138,056,375	₩ -	₩ 2,300,380
Bonds	219,917,877	226,028,338	121,881,021	50,994,203	53,153,114
Long-term borrowings	623,139,402	657,162,045	126,608,448	121,396,433	409,157,164
	<u>₩ 983,414,034</u>	<u>₩ 1,023,547,138</u>	<u>₩ 386,545,844</u>	<u>₩ 172,390,636</u>	<u>₩ 464,610,658</u>

December 31, 2015					
	Carrying amount	Contractual cash flows	Less than six months	6-12 months	More than one year
(In thousands of Korean won)					
Trade and other payables	₩ 58,103,465	₩ 58,103,465	₩ 56,193,891	₩ -	₩ 1,909,574
Short-term borrowings	10,109,490	10,196,369	10,196,369	-	-
Bonds	50,000,000	56,954,000	869,250	869,250	55,215,500
Long-term borrowings	631,792,000	696,008,682	10,110,587	10,165,771	675,732,324
	<u>₩ 750,004,955</u>	<u>₩ 821,262,516</u>	<u>₩ 77,370,097</u>	<u>₩ 11,035,021</u>	<u>₩ 732,857,398</u>

(4) Currency risk

The Company is exposed to currency risk on sales, purchases and borrowings. The currencies in which these transactions primarily are denominated are USD, etc. The Company establishes a plan to manage currency risk through currency forwards, etc., if considered necessary.

1) Exposure to currency risk

The Company's exposure to foreign currency risk as of December 31, 2016 and 2015, are as follows (based on national amounts):

December 31, 2016					
	USD	EUR	CHF		
(In thousands of Korean won)					
Trade and other receivables	₩ 64,553,873	₩ 22,687	₩ -		
Short-term and long-term borrowings	132,339,402	-	-		
Trade and other payables	₩ 1,649,147	₩ 521,643	₩ 1,469		
December 31, 2015					
	USD	EUR	GBP		
(In thousands of Korean won)					
Cash and cash equivalents	₩ -	₩ -	₩ -		
Trade and other receivables	35,037,186	-	10,049		
Short-term and long-term borrowings	110,901,490	-	-		
Trade and other payables	₩ 5,736,321	₩ 52,371	₩ -		

2) The following significant exchange rates applied as of December 31, 2016 and 2015:

	Average rate		Reporting date spot rate	
	2016	2015	2016	2015
	(Korean won)			
USD	₩ 1,160.50	₩ 1,131.49	₩ 1,208.50	₩ 1,172.00
EUR	1,283.30	1,255.16	1,267.60	1,280.53
GBP	1,572.18	1,729.79	1,480.17	1,735.91
CHF	1,177.36	1,175.58	1,181.33	1,185.39

3) Sensitivity analysis

A strengthening (weakening) of the won, a change of 10% in foreign currency exchange rate as indicated below, against the GBP, EUR, CHF and USD as of December 31, 2016 and 2015, would have increased (decreased) profit or loss (before tax) by the amounts shown below. This analysis is based on foreign currency exchange rate variances that the Company considered to be reasonably possible at the reporting date.

	December 31, 2016		December 31, 2015	
	10% increase	10% decrease	10% increase	10% decrease
	(In thousands of Korean won)			
USD	₩ (6,943,468)	₩ 6,943,468	₩ (8,160,063)	₩ 8,160,063
EUR	(49,896)	49,896	(5,237)	5,237
GBP	-	-	1,005	(1,005)
CHF	(147)	147	-	-
	₩ (6,993,511)	₩ 6,993,511	₩ (8,164,295)	₩ 8,164,295

(5) Interest rate risk

The Company is exposed to interest rate risk because they raised a fund through some floating rate liability. The effect of changes in interest rates at 0.5% interest rate is applied to the profit and loss (before tax reflected) as of December 31, 2016 and 2015, are as follows:

	December 31, 2016		December 31, 2015	
	0.5% increase	0.5% decrease	0.5% increase	0.5% decrease
	(In thousands of Korean won)			
Interest expenses	₩ (423,265)	₩ 423,265	₩ (391,168)	₩ 391,168

(6) Capital management

The Company's objectives when managing capital are to safeguard the Company's ability to continue as a going concern in order to provide returns for shareholders and benefits for other stakeholders and to maintain an optimal capital structure to reduce the cost of capital.

In order to maintain or adjust the capital structure, the Company may adjust the amount of return capital to shareholders, issue new shares or sell assets to reduce debt.

The debt-to-equity percentages and net borrowing percentages as of December 31, 2016 and 2015, are as follows:

	December 31, 2016	December 31, 2015
	(In thousands of Korean won)	
Total liabilities (A)	₩ 3,450,636,647	₩ 3,185,659,558
Total equity (B)	4,082,379,460	2,774,834,159
Cash and cash equivalents (C)	230,337,601	34,510,062
Total borrowings (D)	₩ 843,057,279	₩ 691,901,490
Debt to equity (A/B)	84.53%	114.81%
Net borrowing ((D-C)/B)	15.01%	23.69%